

D&C Performance Brief

Mechanical Services

For Skygate Play – Brisbane Airport
At The Circuit, Brisbane Airport

SPECIFICATION
M25-0437

ISSUE
B

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04-02-2026

ARCHITECT
Archipelago



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1. PRELIMINARIES

1.1 Project Description

This project consists of an Area 51 and Game Over.

1.2 Delivery Method – Design & Construct

The mechanical services in this project is to be delivered as a “Design & Construct” procurement. It is the complete responsibility of the contractor to design and construct the works in accordance with the requirements of this document, the Tenant Brief, the Brisbane Airport Corporation Design Guidelines, all mandatory codes and standards, federal act and regulations, and all requirements of the head contract documents.

The contractor shall make all allowances for complete and operational mechanical services installation, including professional fees associated with design and certification, materials, labour, off site costs, on site costs, site allowances, union/EBA allowances, profit and overheads.

1.3 Tender Forms

The Tender Forms attached to this specification, shall be completed, and submitted as part of the Tender.

1.4 Contingency, Provisional and Prime Cost Sums

The amounts stated in the Tender Form shall be included in the Contract Sum. Disbursement of the amounts shall be at the sole discretion of the Superintendent. Any part of the amounts not expended shall be deducted from the Contract Sum.

A Provisional Sum is for foreseeable work which is not detailed in the contract documents.

A Prime Cost Sum is for supply and/or installation of a foreseeable “off the shelf” item which is not detailed in the contract documents.

1.5 Defects Liability Period

Warrant the entire works of the Contract for the Defects Liability Period, which shall be for the period stated in the Head Contract, or if not stated, shall be 12 months from date of Practical Completion.

2. SCOPE

2.1 General

The work carried out shall comprise the manufacture, supply, delivery, installation, testing, commissioning, and maintenance of the Mechanical Services for this project.

All work shall be carried out in accordance with the specification, drawings and such details and instructions as are issued by the Superintendent or his representative during the progress of the work.

This specification shall be read in conjunction with the drawings, all of which are intended to be mutually explanatory. All work called for by the one even if not by the other, shall be fully executed.

The drawings and brief are intended to show the concept and do not necessarily show all equipment required to complete the whole mechanical works. The documents are intended to so the scope to allow competitive tenders to be called.

Where sizes, quantities, capacities, etc. are used in these documents, they are provided to assist in the tendering process only. The contractor is responsible for ensuring that selections final sizes, quantities, capacities, etc. meet the design intent for this project.

The Contractor shall be fully responsible for the complete design of the Mechanical and Electrical Services installation and shall provide calculations and drawings and other documentation as necessary to demonstrate conformance with the design parameters, industry practice, codes, regulations, and standards.

2.2 Extent of Work

The work includes but is not limited to the following main elements:

- Design, documentation and certification of the complete mechanical services by a RPEQ engineer
- 4 Star Greenstar initiatives (not certification) is being targeted for this building, as such the design and construction of the electrical services shall align with the selected points table as issued by the Greenstar Professional.
- Reverse cycle packaged air conditioning systems with economy cycle and carbon dioxide outside air control
- 100% outside air packaged supply air system to serve Snow World.
- Ventilation systems
- Kitchen exhaust systems with make-up air.
- Filters
- Air distribution system as indicated, including provision of all external louvres
- Thermal Insulation
- Electrical works
- Building management system (BMS) to control air conditioning and ventilation systems.
- Interface between BMS and BAC's network.

- Variable speed drives (VSD)
- Mechanical services control panels and switchboards
- Over flashing to external wall penetrations for the mechanical services
- Acoustic treatment to mechanical services to achieve noise levels required by the acoustic engineer, including wall and ceiling penetrations.
- All associated electrical power and control wiring and equipment to facilitate automatic operation of equipment.
- Fire shutdown of nominated equipment via an alarm signal provided by Fire Services
- Provision of workshop drawings in time to enable design and coordination of trusses and other building elements to accommodate mechanical systems
- Liaise with the hydraulic, electrical and fire services to ensure all interfaces between trades have been correctly tendered
- All transport to site, positioning and hoisting of equipment
- Instruction of the Client in the operation and use of the mechanical services systems
- All equipment testing, commissioning and maintenance for the specified period after the Date of Practical Completion
- Allow for final rebalance and recommissioning of air conditioning post lessee fit-out works.
- Design and As Installed drawings to be modelled in Revit to BAC Standards
- Supply of operating and maintenance manuals and "as installed" drawings
- Labelling of all plant and systems
- Twelve months service warranty maintenance period
- Any additional work necessary to provide a complete, operative installation in accordance with performance requirements and miscellaneous work as specified or shown on the drawings
- Certification of the installation

2.3 Work in Other Trades

Related work in connection with the Mechanical Services (performed by other trades) includes the following:

2.3.1 By Main Contractor/Builder

- Concrete plinths, plant decks and enclosures for the mechanical plant
- Refrigeration equipment for Snow World area
- Acoustic walls / barriers
- Concrete plinths shall be 100mm high for mechanical equipment (packaged units, floor mounted fans, MSSB's)
- Access to the roof space and ceiling space for servicing of plant and equipment
- Drain points for condensate waste
- Space in the roof for equipment and ductwork installation
- Design construction and coordination of roof trusses arrangement to accommodate mechanical ductwork and systems
- Acoustic and thermal insulation to roof, walls, and plant decks

- Installation only of door grilles, supplied by Mechanical Services and undercutting doors as shown in drawings
- Formed and trimmed openings through walls, floors, windows, ceilings, and roof for mechanical and electrical components
- Provision of core holes through existing floors and walls
- Under flashing of roof penetrations
- Temporary light and power during construction

2.3.2 By Electrical Services

The following works associated with the Mechanical Services installation are specified in the Electrical Services:

Three (3) phase and single-phase power supplies shall be provided to or adjacent to the mechanical services plant listed below. Cables shall be connected direct to the larger items of equipment and switchboards or to local isolators as noted:

- Mechanical services switchboards – direct connection
 - MSSB-1 - non-essential supply
 - MSSB-2 – non-essential supply

2.3.3 By Fire Protection Services

- Cable to the mechanical services equipment and junction boxes for final connection by Mechanical Services for shut down of ducted air conditioning fan coil units in the event of a fire alarm.

2.3.4 Interface with BAC Communication Network

The requirements for BAC to monitor the Building Management System (BMS) in the building are:-

- Installation of MSSB mounted network switchgear - by Electrical Contractor.
 - The rack will accommodate an Ethernet switch and patch panel for the fibre - by Electrical Contractor.
 - A 12 core fibre cable is to be installed in conduit between the rack and the BAC substation - by Electrical Contractor.
- Cat 6 cables from the rack switch to the air conditioning control panel - by Electrical Contractor
 - One to be connected to the DDC control/communications module.
 - One to be terminated on a wall mounted RJ45 outlet inside the board.
- One RJ45 outlet inside air conditioning control panel - by Mechanical Contractor.
 - Final termination of Cat 6 cable to RJ45 in air conditioning control panel - by Mechanical Contractor.

2.4 Drawings

The following drawings form part of this specification:

Drawing Number	Title
25021-ASH-XX-XX-DR-ME-001001	Locality Plan, Legend & Drawing List
25021-ASH-XX-XX-DR-ME-001100	Ground Floor – Air Conditioning and Ventilation Layout 1
25021-ASH-XX-XX-DR-ME-001200	Ground Floor – Air Conditioning and Ventilation Layout 2

The above drawings shall be read in conjunction with associated architectural, engineering and services drawings. Obtain any additional information or plans considered necessary for a complete understanding of the project.

3. GENERAL REQUIREMENTS

3.1 General

This section outlines general requirements which are applicable to all sections of this specification and the works.

3.2 Standards and Regulations

Conform to the latest editions of all standards and regulations. The following standards and regulations are always applicable:

NCC / BCA	National Construction Code 2022
AS1657	Fixed platforms, walkways, stairways, and ladders — Design, construction and installation
AS1668.1	The use of ventilation and air-conditioning in buildings Part 1: Fire and smoke control in buildings.
AS1668.2	The use of ventilation and Airconditioning in buildings Part 2: Mechanical ventilation in buildings
AS 817	Refrigerants—Designation and safety classification
AS 5149	Refrigerating systems and heat pumps—Safety and environmental requirements
AS3000	Wiring Rules
AS3012	Electrical installations - Construction and demolition sites
AS3666	Air Handling and Water Systems of Buildings - Microbial Control
AS/NZS3823	Performance of electrical appliances – Air conditioners and heat pumps
AS1170.2	Cyclone fixings
AS1470	Health and Safety at Work – Principles and Practices
AS2865	Confined Spaces
AS4836	Safe working on or near low-voltage electrical installations and equipment
- Ozone Protection and Synthetic Greenhouse Gas Management Act 1989 (as amended in 2003) and the Ozone Protection and Synthetic Greenhouse Gas Management Regulations 1995	
- Refrigerant handling code of practice 2007, Part 2 Systems other than self-contained low charge systems	
- All other relevant Acts and Regulations, Local Authority Requirements, Australian Standards and Codes having jurisdiction.	

3.3 Energy Efficiency

The air conditioning and ventilation systems in Class 2 to Class 9 buildings shall be designed and installed to comply with the following clauses of the National Construction Code 2022 to meet the energy efficiency requirements:

■ Part J6 Air-conditioning and Ventilation Systems

This section of the NCC includes requirements for system control, duct insulation, power input for fans, time switch, heating and chilling systems and requirements for miscellaneous exhaust systems.

3.3.1.1 BAC Sustainable Design

BAC strive to achieve sustainable design targets greater than NCC requirements. The building is to have an energy consumption 10% lower than one built to the NCC. The targets also include items such as reduced waste to landfill, clean air and acoustic comfort. Refer to Airport Design Guidelines ADG-020 Sustainable Design.

3.4 BAC Guidelines and Technical Specifications

The design and installation are to comply with the current version for the Brisbane Airport Corporation's (BAC) suite of Airport Design Guidelines and Technical Specifications.

Reference	Title
ADG-020	Sustainable Design
ADG-050	Project Completion Documentation
ADG-051	Commissioning
ADG-601	Electrical General
ADG-620	Mechanical
ADG-625	Building Management Systems
ADG-903	BIM and CAD Standards

3.5 Refrigerant Leakage

Ensure refrigerant leakage does not exceed the maximum value allowed by AS5149 in each room. Refrigerant volumes shall include both the outdoor unit's initial charge, plus additional charge to allow for the installed pipe runs.

In the event the air conditioning system does not comply with this maximum quantity of leakage, provide an appropriate method of leak detection or removal to meet the requirements of the standard and the Certifier.

3.6 Quality of Materials and Workmanship

All materials shall be new, of the best quality and of the class most suitable for the purpose specified.

All equipment shall have a verified history of successful use in the commercial environment. No prototype equipment will be accepted.

All work on the site shall be supervised by a competent tradesman supervisor who shall be present on site when requested by the Superintendent, for all phases of the project. A representative shall be nominated, who shall be present and responsible for the day to day running of the project.

The entire mechanical installation shall be executed by licensed, experienced tradesmen, and shall be completed in a neat and first-class tradesman-like manner to the satisfaction of the Superintendent.

3.7 Uniformity

All fittings, accessories and equipment of the same type shall be of the same manufacture and catalogue number.

3.8 Manufacturer's Warranty

Provide manufacturer's warranty agreements where applicable. Complete all necessary documentation on the Owner's behalf. Include copies of warranties in the maintenance manuals.

3.9 Nominated Suppliers

Where any item or component is nominated in this specification or on the drawings by the name of the manufacturer, it is expected that such items or components will be used. The expressions "or equal" or "or similar" allow the substitution of components with the approval of the Superintendent. Any request for such approval must be accompanied by supporting information such as improved delivery, superior performance, statutory requirements, capital cost savings, etc.

Where any item or component is nominated in this specification by the name of the manufacturer, it is expected that the item complies with the clauses of the specification regardless of whether it is a standard item as manufactured.

3.10 Submissions

3.10.1 General

Submit drawings, samples, technical data, or other details as applicable for systems as specified prior to ordering or installation.

Allow sufficient time for checking and return of submissions. Co-ordinate submissions of related items. This drawing review will not relieve the Contractor of his overall responsibility for compliance with the Specification, Standards and Authorities and for the proper co-ordination, installation, and satisfactory operation of the services in accordance with the specification and drawings.

Should re-examination of shop drawings be deemed necessary due to inefficiencies of the Contractor, additional checking shall be at the Contractor's expense at applicable rates.

3.10.2 Drawings

Provide copies of "shop drawings" and "construction drawings". Drawings shall be produced as CAD and PDF's and show the works required for the project. Drawings shall be a scale of 1:50.

3.10.2.1 General

- Site works.
- Associated building works requirements.
- Penetrations through structural and/or fire/smoke rated walls, floors, ceilings, and the like.

- Layouts and details of all plant and equipment.
- Schematic electrical and control diagrams.
- Switchboards.
- Access locations.
- Dampers.
- Air flow rates.
- Schedules
- Relevant details of each assembly, component, and connection.
- Information relative to fabrication, surface treatment, transport, and erection.
- Submit 3-dimensional shop drawings including reference grids in XYZ coordinates.
- Shop drawings submission medium: IFC2x3, DWG(3D) & PDF.

3.10.2.2 Ductwork

- duct numbering
- duct heights
- duct insulation types

3.10.2.3 Pipework

- pipe insulation types

3.10.3 Samples

Prior to placing of orders for fittings, equipment, accessories, and the like, obtain a list of sample items and details for approval to be reviewed by the Superintendent.

Allow reasonable time for the delivery and review process to avoid unnecessary delays in the work program.

3.10.4 Technical Data

Provide manufacturers technical specifications, type test report, performance data, installation details etc. as required.

3.10.5 Co-ordination

Co-ordinate shop drawing design with other building and services elements.

3.11 Certification

Certify that the works have been installed to the requirements of all applicable rules and regulations.

Airports Act 1996, Airports (Building Control) Regulation 1996
Design Certificate
Installation Certificate
Inspection Certificate

3.12 Hazard Assessment and Risk Management

Comply with the requirements of the Workplace Health and Safety Act and Regulations with respect to hazard identification and risk management.

3.13 Electromagnetic Compatibility

Ensure that the entire installation covered by this specification meets the electromagnetic compatibility requirements of the Australian Communications and Media Authority.

3.14 Interference

The installation, equipment and/or appliances shall be such that it does not cause any interference with, nor should it be affected by radio, television, communication, or other electrical equipment. In the event of the inherent characteristics of the electrical installation being such that interference will occur, provide efficient devices capable of eliminating the interference at no additional cost.

3.15 Site Conditions and Precautions

The site will be occupied and operational for the duration of the contract. Ensure that systems remain fully connected and operational at all times, except where shutdowns are required.

Where shutdowns of systems are required:

- Provide the Architect and Owner with written notification at least 5 days in advance of the intended shutdown time.
- Keep each shutdown time to a minimum.
- For each shutdown, complete the works and ensure that the systems are fully operational prior to leaving the site for the day.

Carry out the work in such a manner as to interfere as little as possible with other trades and persons having access rights. Ensure that normal access ways within the buildings are unobstructed.

3.16 Access to Site

Generally, access to the site, working hours and storage facilities shall be agreed with the Project Manager at the pre-contract site meeting. However, certain restrictions may be applicable to the project and the Contractor shall include within his tender for any restrictions on access, working hours that may be imposed by the Local Authorities.

3.17 Packing Storage and Protection

Any items suffering damage in transit or whilst on site shall be rejected and replaced without cost to the Contract.

- All plant, equipment, apparatus, materials, and parts shall be delivered to the site in a new condition and properly packed and protected against damage due to handling or due to adverse weather or other circumstances and so far, as practicable shall be kept in the packing cases or under protective covering until required for use.

- The Contractor shall arrange suitable storage areas for materials, plant, and equipment, as necessary. Storage shall be on purpose made racks, mounts, etc., to ensure no damage or deformation of items. Plant and equipment should preferably be placed in final location, and should not be brought on to site before the area is ready, subject to access requirements. All plant and equipment brought on to site whether in final location or not shall be protected as described below.
- The Contractor shall suitably protect, encase, cover over etc., as may be appropriate all plant and equipment, instruments, pipework insulation etc. installed by him against damage due to building operations or adverse weather or other causes up to date of completion and shall make good any damage and hand over the entire installation in a new and undamaged condition. This shall apply to fixed and unfixed materials, plant, equipment etc.

3.18 Witness and Hold Points

Witness Points can be performed via photographic evidence only, precluding the requirement for a physical witnessing of the item detailed.

Photographs are to be labelled with the item name and specific location and provided in jpg format.

- a. Witness points – Via photographs if agreed prior with BAC
 - i. Formal written acceptance including all identified issues provided 3 business days following site inspection
- b. Witness points – Requiring site inspection (if required)
 - i. 10 business days notice required to BAC prior to physical inspection.
 - ii. Formal written acceptance including all identified issues provided within 2 business days following site inspection
- c. Hold Points – Requiring site inspection
 - i. 10 business days notice required to BAC prior to inspection.
 - ii. Formal written acceptance including all identified issues provided within 2 business days following site inspection

Refer to BAC Airport Design Guidelines ADG-620 for construction witness and hold points.

3.19 Protection Guards

Moving parts of all machines where exposed, such as fan drives, pump drives, etc., shall be protected by wire mesh guards or totally enclosed guards in accordance with the requirements of the Health and Safety at Works Act.

3.20 Coordination with other trades

The positioning of equipment as shown on the drawings is approximate and diagrammatic only, and these drawings shall not be used for exact setting out. Locations shall be confirmed prior to installation.

Coordinate with the building structure (e.g., columns, beams, trusses).

Coordinate with electrical services for electrical supplies to the as-installed equipment. Coordination shall include (but not limited to):

- Circuit breaker sizes

- Cable sizes
- Sub-main sizes
- Maximum electrical demand

Coordinate with hydraulic services for location and size of drainage points.

3.21 Installation

Install equipment and services square and in line with the building elements, plumb and at consistent mounting heights.

Where possible fix directly to the building structure.

Provide noggings, brackets, fixings and supports to manufacturer's recommendations and as required.

Where timber is used for supports or noggings provide Cyprus pine or similar termite resistant species.

Organise reticulated services neatly and with applicable segregation.

Provide separate cable support systems for each reticulated service and support all cables over their entire route length.

Provide all access equipment and temporary lighting necessary for the installation of mechanical services.

3.22 Seismic Restraint

Provide equipment and equipment installation suitable for seismic restraint as required by NCC/BCA 2022 Sections B1P1 and B1P2. The reference standard is AS1170.4 Earthquake actions in Australia, and the most relevant section is Section 8 Design of Parts and Components.

For this specification the seismic restraint pertains in particular to:

- Support brackets
- Fixings and fasteners, especially fixings into concrete and masonry
- Ductwork especially long duct runs where differential building movement could cause damage
- Heavy equipment, especially where installed at high level relative to ground level, where installed overhead and where installed in ceiling spaces
- Pipe runs subject to differential movement of building elements
- Fixings for electrical switchboards
- Equipment associated with safety systems and smoke control systems
- Equipment and plant mounted on vibration isolating mounts

The Mechanical Services shall provide a Form 15 design certification of Seismic restraint design for the project, signed by a suitably qualified Structural engineer (RPEQ).

Upon completion of the project, provide a Form 12 Installation certificate, signed by a suitably qualified Structural engineer (RPEQ).

All associated costs for Certification (including design, shop drawings and site inspections) are to be included in the Mechanical Tender.

3.23 Building Penetrations

Maintain integrity of building systems such as waterproofing and termite management at building penetrations.

3.24 Maintenance Access

Locate all relevant equipment to ensure adequate maintenance access in accordance with Workplace Health and Safety regulations.

3.25 Corrosion Protection

All metallic materials and components shall be suitably protected against corrosion.

Bolts, screws, rivets, etc. for external use shall be non-ferrous or stainless steel, or where approved galvanised steel.

Brackets, plinths, rods, etc. shall be hot dip galvanised after fabrication.

All steel sheets shall be corrosion protected by means of degreasing, priming, and painting as a minimum.

3.26 Corrosion Protection in Highly Corrosive Environments

As the site is in a highly corrosive environment, the contractor shall allow for necessary corrosion protection over the above the standard to ensure longevity of plant and equipment, using specialist suppliers such as Blygold Australia Ltd.

3.27 Finishes

All equipment shall be painted unless the equipment material is chromium, anodised aluminium, plastics, stainless steel, or other non-ferrous metals.

Paint and paint application shall comply with AS/NZS 2311 and AS/NZS 2312.
Carry out all powder coating to AS3715.

3.28 Adhesives

Provide adhesives and sealants capable of transmitting imposed loads, sufficient to ensure the rigidity of the assembly or the integrity of the joint and which will not cause discolouration or lack of adhesion of finished surfaces.

3.29 Fasteners

Provide masonry anchors, bolts, nuts, washers, screws, nails, and plugs as required for installation of mechanical services.

Fastener materials, finishes and physical sizes shall be as appropriate for the loads, the types of materials in contact with the fastener and the environment.

3.30 Metals

Provide metal sections and sheet as required for the mechanical services.

Finish invisible joints by welding, brazing, soldering, grinding, and buffing prior to application of finishes such as galvanising or painting.

Repair galvanised or electroplated metals using two pack organic primers to AS/NZS 3750.9.

3.31 Labelling

Provide labels to equipment as required for identification, operation, and warnings.

Labels shall be Traffolyte.

Labels shall be of proven longevity without significant deterioration.

Externally mounted labels shall be weather and corrosion resistant.

Labels shall be fixed as follows:

- Internal – adhesive (double sided tape, liquid nails) or screws
- External – stainless screws

Externally mounted fixings shall be corrosion resistant.

Warning labels shall be white lettering on red background with minimum 10 mm high lettering.

Minimum letter heights shall be 3mm.

Supply and install labels to:

- Condensers
- Indoor AC Units
- ERV's
- Fans
- Controllers / Switches
- Sensors (Temperature, Humidity, Pressure, Airflow)
- Ductwork indicating direction of airflow (SA, RA, EA, OA)

3.32 Operation and Maintenance Manuals

Provide Operation and Maintenance Manuals for all systems and components covered by this specification. Include As-Installed documentation for the entire installation in these manuals.

Provide electronic copies of manuals and drawings which are permanently labelled with the Project Name, Contract Description and Contractor's Name.

Include the following in each set of manuals:

- Index.
- Copies of all test results, certificates, and approvals.
- Manufacturer's operating instructions and details of model numbers, serial numbers, operating instructions, and manufacturer's guarantee certificate for equipment and appliances.
- Brief description of each component of the installation.
- A recommended maintenance programme for all equipment installed.
- A recommended list of spare parts.
- Log sheets to record maintenance activity and relevant dates.
- Addresses, names and phone numbers of all relevant suppliers, spare parts suppliers, service companies etc.
- As-installed drawings as follows:
 - Layout drawings in the same size and form as the workshop drawings but modified to reflect all changes which occurred during construction, and access locations, position of dampers, air quantities and locations of all registers shown.
 - System schematic diagrams and schedules.
 - The title block of each drawing amended to include Contractor's name and contact details, Contractor's drawing number and date of issue.
 - In Revit RVT or IFC format.
 - In AutoCAD DWG format.
 - In Adobe Portable Document format (PDF).

Provide a draft set of the Operation and Maintenance Manuals to the Superintendent not more than 2 weeks before the date of practical completion, for the purpose of review and comments.

Submit final sets of the Operation and Maintenance Manuals to the Superintendent not more than 2 weeks after the receipt of comments on the draft from Superintendent.

3.33 Clean Up

At completion clean all mechanical equipment to as-new condition. Remove all waste and debris on a continuous basis.

3.34 Testing and Commissioning

Fully test and commission the complete mechanical installation prior to practical completion. Record all test results and submit for inspection prior to practical completion.

3.35 Test Instrument Calibration

All test instruments shall have current calibration certificates. Certificates shall be shown to the Superintendent prior to commencement of each test. Certificate currency shall be 12 months maximum.

3.36 Inspection at Practical Completion

Arrange for a Practical Completion Inspection by the Superintendent. A minimum of 5 working days' notice shall be given. Ensure that the installation is fully operable and practically completed before requesting an inspection.

Ensure that the installation is fully operable and practically completed before requesting a Practical Completion inspection.

The Superintendent or his nominated representative is not regarded as the Contractor's checking agent.

Should the inspection reveal that the installation is not fully operable or practically completed and require further inspections, these inspections will be charged to the Contractor.

All defects and outstanding works shall be rectified within fourteen (14) days.

3.37 Instructions and Training

Provide skilled operators to instruct such persons as may be nominated by the Superintendent in the effective operation of all the mechanical systems in the installation. Instructions shall not be given until completion of all tests and adjustments.

Conduct training at agreed times and locations and for durations commensurate with the complexity of the systems.

3.38 Maintenance and Warranty during Defects Liability Period

Carry out preventative maintenance on a regular schedule for systems as detailed in other sections of this specification.

Promptly rectify all defects during the defects liability period. Attend site within 1 working day of notification of defect. Critical system defects shall be rectified in the time schedule as detailed in other sections of this specification.

4. DESIGN CRITERIA

4.1 Performance Guarantees

4.1.1 Capacities

The capacities of all items of equipment provided shall be not less than as nominated by the specification, equipment schedules and drawings. The design is based on equipment complying with NCC (National Construction Code 2022) energy requirements.

Guarantee that the required inside conditions shall be achieved under the design criteria stated below. The capacities listed, however, are minimum capacities for the design inside and outside conditions stated.

4.1.2 Performance

The air quantities and noise levels shall be obtained at all locations as required by the drawings and specification.

4.1.3 Operating Duty

The Contractor shall be responsible for the correct selection of all equipment and power of driving motors and for any other such design requirements which affect system performance.

Static pressures (for fans) noted in Equipment schedules for Tender purposes are “indicative only” and the Contractor shall determine actual static pressures from “shop drawings”. Contractor to submit calculations for review.

4.1.3.1 Fans

Fans shall be selected on the following speeds @ 50Hz:

- 1500 rpm (4 pole)

Fans shall be selected with the following spare capacity:

- 10% flow, 20% pressure

4.2 Air Conditioning Systems Design Criteria

The total equipment capacity has been determined using the following design criteria, supplemented as necessary by examination of the working drawings of the building, which are available for examination by the tenderers.

Item	Requirement	Comment
People	70 W/person sensible 60 W/person latent	Occupant heat emissions are dependent on activity. Increase if required for actual activity and resultant metabolic rate.
Outside air	7.5 L/s.person 10 L/s.person for Snow World	With suitable particulate filtration compliant with AS1668.2
Building fabric	As per architectural design	If unknown for new buildings assume BCA Part J DtS.
Building glazing	As per architectural design	If unknown for new buildings assume BCA Part J DtS.

Building sealing and infiltration	Areas with openable doors to outside: 1.0 ACH infiltration Areas with external windows: 0.5 ACH infiltration Internal areas: 0 ACH infiltration	If building air tightness is pressure tested calculate the expected infiltration at an external wind pressure of 50Pa. (50Pa is equivalent to 9.2m/s wind speed which is approximately the 99% percentile wind speed.)
Fan heat gain	To be calculated	Based on fan motor power and efficiency
Duct heat gain	To be calculated	Based on duct runs, insulation, and temperatures
Duct leakage	5% maximum	AS4254 compliant Systems >3000L/s to be pressure tested (as per AS4254 compliance requirements) and sealed with aerosolised vinyl acetate polymer if testing fails
Heat load contingency	Space cooling sensible: 10% Space cooling latent: 10% Space heating: 10%	To be applied to calculate space cooling and heating loads
Supply air temperature	Public spaces: 13.0°C Admin with VAV systems: 11.0°C	Air temperature out of supply air grilles serving the air-conditioned space (ie not coil-off temperature) Low temperature VAV system, adjust as required if airflow at design loads is <2.5 L/s.m²
Hours of operation and schedules	HVAC systems shall be designed to operate at any time of the day or night	For energy analyses confirm operation hours and schedules with BAC

4.2.1 External Design Criteria

Outdoor design criteria are nominated in BAC Design Guidelines and shown in the following tables:

DESIGN TEMPERATURE: COMFORT			
Site	Summer		Winter
	[°C DB]	[°C WB]	[°C DB]
Brisbane Airport	32.0	25.6	7.0

These conditions will be applied to all relevant areas.

External heat rejection plant such as condensers will be specified for an air on temperature as advised below.

AMBIENT DESIGN TEMPERATURES: Air-on condition for sizing condensers		
Site	Summer	Winter
	[°C DB]	[°C DB]
Brisbane Airport	35.0	5.0

Extreme conditions which the whole system shall be capable of safe and stable control and operation at reduced capacity as advised below.

AMBIENT DESIGN TEMPERATURES: Air-on condition for extreme operation		
Site	Summer	Winter
	[°C DB]	[°C DB]
Brisbane Airport	45.0	-5.0

4.2.2 Internal Design Criteria

Internal design parameters are detailed below:

Area	Summer	Winter
Air Conditioned Areas (general)	24.0°C DB 60% RH*	21.0°C DB
Snow World	-8°C DB	-8°C

* No active humidity control will be provided. Air conditioning plant will be designed to limit humidity to 65% under design ambient conditions. Selection of plant will provide inherent dehumidification to achieve the above, and no specific control of humidity will be applied.

4.3 Climatic Zone

The project is in the following NCC Climatic Zone in accordance with the NCC

Location	Climate Zone (NCC)
Brisbane	2

4.4 Design Life

Element	Minimum Design Life
Air conditioning unit – air cooled (<10kW)	10 years
Air conditioning unit – air cooled (>10kW)	15 years
BMS	20 years
Ductwork and fittings	30 years
Electrical	Refer BAC Design Guidelines ADG-601
Fans	20 years

4.5 Indoor Noise Criteria

Mechanical systems will be designed and selected to maintain internal noise levels below the levels, in accordance with the Acoustic Consultant's recommendations.

In the absence of briefed requirements, levels listed in AS 2107 "Acoustics - Recommended design sound levels and reverberation times for building interiors" will be used.

Acoustically treated transfer ducts will be utilised where required.

Air conditioning units will be selected on medium fan speed to meet internal noise levels.

4.5.1 Acoustic Design Criteria

The noise levels specified below form the basis of the acoustics on this project and should be maintained under all conditions of operation of the mechanical systems. Ensure that the noise levels in all occupied spaces due to the mechanical equipment shall not exceed the scheduled areas below.

<u>Acoustic Design Criteria Space</u>	<u>Design Sound Levels L_{Aeq} dB(A)</u>	
	<u>Satisfactory</u>	<u>Maximum</u>
Corridors & Lobbies	40	50
Kitchen Areas	45	55
Staff Common Rooms	40	45
Toilet / Change / Showers	45	55
General Areas	40	45
Private Offices	35	40

The NR levels should be determined by measuring the 60 second fast response octave band L_{eq} levels in the respective octave bands. Background level measurements shall also be recorded in each octave band and the tests carried out at a time when the background levels are at least 5 dB below the required criteria levels. Levels are measured no less than 1m from the floor or 1.5m from any wall.

The sound level meter shall be coupled to an octave band analyser complying with the SAA Standards

5. AIR CONDITIONING UNITS

5.1 General

Supply and install new air conditioning units of the most recent design and the most energy efficient type available from the manufacturer.

Refrigerant for all air conditioning units shall have zero Ozone Depletion Potential, low Global Warming Potential, short Estimated Atmospheric Life, and low Total Equivalent Warming Impact.

Unless specified otherwise, units shall provide the specified minimum capacity, when tested in accordance with AS 3823.1 standard rating condition T1 (cooling) and H1 (heating).

Units must be selected with an air on condenser temperature in accordance with section 4.2.1 *External Design Criteria*.

The units shall be complete with all necessary internal operating and safety controls.

Each unit shall be quiet in operation and shall comply with all Code and Authority requirements for sound levels. Fan coil units shall be selected to ensure that noise levels specified for the occupied spaces are not exceeded.

Units which are excessively noisy or transmit excessive vibration will be rejected.

Both evaporator and condenser coils shall be of aluminium fin/copper tube construction.

The condenser coil shall be treated with an approved type chemical coating process to provide a high level of resistance to corrosion without reduction in heat transfer rates or other coil performances. Refer to clause 3.26.

5.2 Standards

The following standards are applicable to this section:

AS 1170.4	Structural design actions, Part 4: Earthquake actions in Australia
AS 1324	Air filters for use in general ventilation and air conditioning
AS 1571	Copper – Seamless tubes for air conditioning and refrigeration
AS/NZS 3000	Wiring rules
AS 3500	Plumbing and drainage (series)
AS 3666	Air handling and water systems in buildings – microbial control
AS/NZS 3823.1.1	Performance of electrical appliances – Air conditioners and heat pumps Part 1.1: Non-ducted air conditioners and heat pumps - Testing and rating for performance
AS/NZS 3823.1.2	Performance of electrical appliances – Air conditioners and heat pumps Part 1.2: Ducted air conditioners and heat pumps - Testing and rating for performance
AS/NZS 3823.2	Performance of electrical appliances – Air conditioners and heat pumps Part 2: Energy labelling and minimum energy performance standards (MEPS) requirements Refrigerant handling code of practice 2007
AS 4041	Pressure Piping
As 4426	Thermal insulation of pipework, ductwork, and equipment – Selection, installation, and finish
AS/NZS 5149.1	Refrigerating systems and heat pumps – safety and environmental requirements, Part 1: Definitions, classification, and selection criteria (ISO 5149-1:2014, MOD)
AS/NZS 5149.2	Refrigerating systems and heat pumps – safety and environmental requirements, Part 2: Design, construction, testing, marking and documentation (ISO 5149-2:2014, MOD)
AS/NZS 5149.3	Refrigerating systems and heat pumps – safety and environmental requirements, Part 3: Installation site (ISO 5149-3:2014, MOD)
AS/NZS 5149.4	Refrigerating systems and heat pumps – safety and environmental requirements, Part 4: Operation, maintenance, repair, and recovery (ISO 5149-5:2014, MOD)

5.3 Energy Performance

All units with a capacity of 65kW or under must have an annual energy efficiency ratio (AEER) which complies with the minimum energy performance standards (MEPS) when tested in accordance with AS/NZS 3823.2.

5.4 Safe Trays

All ducted type air conditioning fan coil units and air handling units, unless plinth-mounted in drained plantrooms, shall be provided with a fully sealed fabricated, galvanised steel safe tray. Trays shall extend 100mm past all valves and flexible connections. The safe tray shall be complete with its own drain. Safe tray and condensate drains shall not be interlinked.

5.5 Condensate drains

Small unitary split air conditioning fan coil unit and safe tray drain pipework shall be uPVC.

Route condensate drains from each air conditioning unit, fan coil unit or similar drain point to waste points as shown on the drawings.

Pipework shall be installed with minimum constant grade of 1 in 100, adequately trapped, and shall comply with AS3666 and local authority requirements. A P-trap shall be used in all non-pumped or indoor components where the drain outlet is exposed to negative air pressure created from the indoor fan to assist drainage unless otherwise indicated by manufacturer's instructions.

Provide a separate condensate drain for each item of plant. Provide a label identifying which unit is served by each condensate drain. The label shall be visible in the tundish.

Drains shall be sized at least one pipe size larger than the unit connection, or as shown on the drawings, whichever is the larger.

Insulate condensate drains using elastomeric closed cell pipe insulation with a minimum of 12mm wall thickness on drains lines. Where practical install without slitting. Where slitting is unavoidable, it shall be glued and taped with foam tape.

5.6 Air Cooled Packaged Air Conditioning Unit

Provide a single piece packaged system air cooled type, reverse cycle units with inverter compressor. The unit shall be horizontal type fitted with a axial condenser fan and consisting of fan coil section and condensing section. Provide anti-vibration mounts under the unit.

The unit shall be housed in a weatherproof cabinet constructed from powder coated or enamelled galvanised steel.

Unit panels shall be fastened using stainless steel bolts and/or screws. Access panels shall be readily removable for service of internal components.

Provide a drained safety tray under the whole unit.

Unit manufacturer shall be:

- Temperzone
- ActronAir or approved equal.

Substitutions will not be accepted unless approved by the Superintendent. Any change to electrical or drain requirements because of any approved substitution will be the responsibility of the contractor.

5.6.1 Fan Coil Section

Units shall be of approved factory prefabricated manufacture and shall comprise of air inlet section, direct expansion coil and fan.

Fan coil section shall have internal foil-faced insulation. Insulation shall be of sufficient thickness to prevent condensation in the environment in which the unit is installed.

Condensate trays shall be stainless steel or marine grade aluminium.

Readily removable panels shall be provided for access to all components and to suit the arrangements shown on the drawing.

Fans shall be mounted on approved vibration isolating mounting.

Provide VSD or EC speed control of fan motors as documented.

5.6.2 Air Cooled Condensing Section

Each condensing section shall comprise hermetic reciprocating or scroll type compressors, axial type condenser fans with moisture resistant motors, and finned tube condenser coil.

Condenser fans shall be of the axial direct driven type selected for low noise levels.

The condensing unit shall be fully weatherproof and lined with suitable acoustic insulation, minimum 13 mm thick. Provide loaded vinyl hood over compressor(s).

Provide head pressure control to reduce the condenser fan speeds whenever the condenser head pressure is down to a suitable level.

Safety features to include:

- HP, LP, and loss of refrigerant
- Anti-rapid cycle timer and internal overload for compressor protection
- Circuit breaker control circuits
- Time and temperature controlled de-ice preventing icing up of outdoor coil during heating cycle.
- Frost protection on cooling cycle
- Sensor fault indication
- Crankcase heaters for compressors
- Compressor minimum run time to ensure oil return
- Service valves on compressors
- Phase failure and phase rotation protection device

- 24-volt control circuit

5.7 Outdoor Air Processing Unit

Equipment offered shall be the product of a commercially established manufacturer of specialised HVAC equipment that has operated for a minimum of 15 years. They shall offer local distribution, customer service and warranty support.

Unit manufacturer shall be:

- Armcor OAP series
- Airchange PMA series or approved equal.

Cabinet

- Cabinets shall be of pentapost construction with aluminium profiles, polymer corner joiners and 50mm, double skin, polyurethane sandwich panels to prevent unit sweating.
- Units shall have internal condensate drain trays for evaporator coils. Provide water traps which run to the nearest existing waste point on site.

Refrigeration System

- Primary compressors shall be BLDC inverter type allowing for stable and highly efficient variable capacity control. Digital scroll compressors or fixed speed compressors fitted with variable speed drives shall not be used.
- Secondary compressors shall be fixed speed twin rotary type.
- Electronic bi-flow expansion valve(s) shall be used with BLDC inverter compressors to ensure proper variable capacity operation.
- Compressor start time delays shall be managed by the manufacturer's integrated control system to prevent compressor short cycling.
- Defrost mode shall be managed by the manufacturer's integrated controls system.
- Compressor speed of BLDC inverter compressors shall be controlled by the manufacturer's integrated control system to ensure safe and reliable operation.
- High and low refrigeration pressure cut offs shall be provided.
- The refrigeration system shall be reverse cycle to allow for heating mode.

Supply Air System

- Supply air fans shall be variable speed electronically commutated (EC) forward curve centrifugal type.
- Ingress protection class of evaporator fans shall be IP44 or higher.
- Fan speed shall be controlled by the manufacturers integrated control system.

Condenser Air System

- Condenser air fans shall be electronically commutated (EC) axial type.
- Condenser air fan blades shall be constructed out of die cast aluminium or glass fibre reinforced composite materials.
- Efficiency class of condenser air fans shall be IE4.

- Ingress protection class of condenser air fans shall be IP54 or higher.
- Condenser air fan speed shall be controlled by the manufacturer's integrated control system and modulate with respect to refrigeration head pressure for stable and efficient operation.

Condenser and Evaporator Coils

- Condenser and evaporator coils shall be constructed of copper tube and aluminium fins.
- Multiple stage refrigeration systems shall have interlaced circuitry coils.
- Evaporator coil face velocity shall not exceed 2m/s for effective dehumidification.
- Test pressure for coils shall be 2200 kPa (22 bar).
- All coils shall include an oven baked resin coating for corrosion protection.
- Condensers shall be remote mounted for a split configuration.

Electrical

- Electrical switchboard(s) shall be factory assembled, internally wired, and housed within a weatherproof enclosure. It will conform to Australian Standards and specific state electrical authorities.
- Electrical cables shall be run neatly in cable duct.
- All wiring shall be independently colour coded.
- Circuit breakers, contactors and overloads shall be factory fitted for all three phase motors.

Controls

The unit shall incorporate the manufacturer's control system to manage the temperature control of the supply air.

Outdoor Air Processing Units

The outdoor air processing units shall have their own onboard controls system. The central controllers shall not control these units.

The units shall be setup to deliver initially a 8°C supply air off temperature.

6. PLINTHS

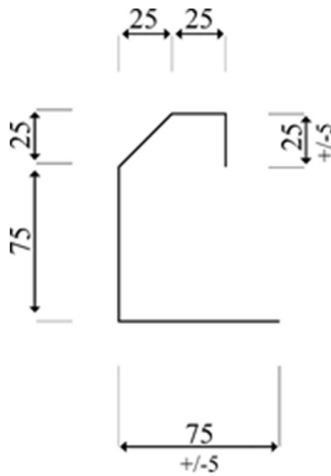
6.1 General

All Mechanical equipment is to be raised off the plant room floor on a 100 mm high plinth. Exposed concrete edges are not acceptable. Form slab using a steel plinth surround with 45-degree chamfered profile. Any mitred corner joints are to be painted with cold gal or similar. Refer to drawings detail for preferred profile.

Provide plinths for the following Mechanical Equipment:

- Ground mounted MSSB's

6.2 Features



Plinth surround Details

7. VENTILATION FANS

7.1 General

7.1.1 Scope

Specified in this section is the minimum requirements for the construction, installation, and testing of the ventilation fans.

7.1.2 General

The fan static pressures and motor kilowatts listed in the Equipment Schedule are provided for guidance purposes only. Check the fan system resistance based on the actual equipment offered. Calculations for the final fan pressure requirements are the responsibility of this contractor. Ensure that the equipment selected complies with NCC Part J6, J6D5 Fans and duct systems.

The fans shall deliver not less than the specified air quantity against the total resistance of the system, as installed.

Fans and motors shall be selected with allowance to provide air flow rates of 10% more than that specified against the corresponding increase in system resistance.

Impellers and other rotating parts shall be balanced after assembly and the fans shall be free of undue vibration at all operating speeds.

Where scheduled, provide an electronic speed controller in the mechanical switchboard or adjacent to the fan (as appropriate) for balancing purposes only. Remove knob when balanced and mount adjacent. Label controller "BALANCE SET, DO NOT TAMPER."

Fans suspended from the roof structure shall be suspended by spring mounted supports complete with neoprene pads and grommets to prevent vibration transmission to the building structure.

All motor/impeller rotors shall be statically and dynamically balanced to International Standard ISO 1940 - "Balance Quality of Rotating Rigid Bodies" and shall be free from vibration at all operating speeds and during starting and stopping cycles.

7.2 Standards

The following standards are applicable to this section:

AS1044	Limits of electromagnetic interference for electrical appliances and equipment
AS1668.1	The use of mechanical ventilation and air conditioning in buildings: Part 1 Fire and smoke control in multi compartment buildings
AG601	Gas Installation Code
NCC	National Construction Code 2022

7.3 In-line Axial Flow Fans

The fans shall be of the long case axial type, of "Fantech", "Pacific Ventilation" or equal and approved.

Impellers shall be aluminium.

The fans shall be suitable for horizontal or vertical mounting.

7.3.1 Casing

Fans shall have flanged hot dipped galvanised tubular casings extending over the length of the fan and motor (unless specified otherwise) complete with an airtight access door suitably placed for easy fan inspection.

7.3.2 Fan Motor

Motor bearings shall be of the ball or roller type, sealed for life or grease packed fitted with lubricant retaining and dust excluding seals extended through the casing, and shall be selected for an operating life of not less than 100,000 hours.

7.3.3 Inlet/Exit Cones

Fans which intake from or discharge to free space or a plenum shall have a manufacturers inlet and/or discharge cone.

Fans shall be of the non-overloading axial aerofoil type of approved manufacture, selected for a minimum efficiency of 65% at the design air flow rate. The aerofoil impeller blades shall have an adjustable pitch.

7.4 In-line Centrifugal and Multiflow Fans

Fans shall be installed in a square galvanised steel housing fitted with flanged duct connections. Impellers shall be either an adjustable pitch mixed flow type or backward curved centrifugal impeller as detailed in the fan schedule.

Electric motors shall be external rotor type when speed adjustment is required. Standard totally enclosed motors are suitable when single speed or two speeds are required.

8. FILTERS

8.1 General

8.1.1 Scope

Specified in this section is the minimum requirements for the construction, installation, and testing of air filters.

The air filters shall be installed in accordance with manufacturer's recommendations complete with all accessories necessary for their proper performance. All filter connections to adjoining equipment, panelling or support framing shall be sealed airtight to prevent any air bypassing the filter media.

Adequate access shall be provided to each filter for inspection, maintenance, and removal.

Provide one spare set of all replacement filter media for installation at the end of the warranty period.

8.2 Standards

The following standards are applicable to this section:

AS1324	Air filters for use in general ventilation and air conditioning
AS1668.2	Mechanical ventilation for acceptable indoor air quality
AS4260	High Efficiency Particulate Air (HEPA) filters

8.3 Channel Slides

Provide channel slides for mounting of filters inside access filter housings. Provide airtight hinged doors for access and removal of filters. Provide resilient gaskets at each end of the filter slides to prevent air bypass.

8.4 Dry Media – Deep Bed Type F5

Minimum performance shall be:

- Make: Camfil (or approved equal)
- Model: Hi-Flo A-Series
- Type: Multi pocket bag filter
- EN 779:2002 Filter class F5
- Economical change point pressure drop 250Pa
- Fire rating UL 900
- Type: 600 deep

8.5 Kitchen Exhaust

The kitchen exhaust systems are to include additional filtration to allow horizontal discharge. The additional filtration, such as UV light, ozone injection or electrostatic filtration is to remove odour and smoke contamination from air stream. The BMS is to monitor the filtration fault alarms.

9. KITCHEN HOOD

9.1 Standards

The following standards are applicable to this section:

AS1324	Air filters for use in general ventilation and air conditioning
AS1668.1	Fire and smoke control in buildings
AS1668.2	Mechanical ventilation in buildings

9.2 General

Provide kitchen exhaust hood(s) as shown in the drawing in accordance with AS1668 and the requirements of the Local Authority.

The hood shall be constructed from 304 stainless steel, from whole sheets with a minimum of joints. All folds shall be machine folded and smooth. All joints shall be welded, ground, and polished. All edges shall be smooth and geometrically true. Rivets showing through the cleanable surface are not permitted.

The hood shall be constructed from sufficient gauge material to ensure both stability and freedom from vibration. Minimum material gauge shall be 1.0mm.

The hood shall be constructed, fitted, and sealed hard against ceilings and walls, to prevent the harbourage of vermin. Install stainless steel trimming angle around hood at ceiling level. Suitably flash around the hood to the ceiling line neatly.

The hood shall be cleaned and polished after installation.

The ductwork shall be in accordance with Section 10, with the additional requirements for kitchen exhaust ductwork and insulation systems as follows:

- Use rectangular radiused bends throughout. Do not use turning vanes or dampers. Use not less than 1.2mm galvanised sheet metal for all ductwork not contained in a fire rated shaft.
- Ductwork and plenum to exhaust system shall be constructed of 1.2mm galvanised metal.
- Horizontal duct runs shall be kept to a minimum and shall be installed with a rise in the direction of airflow of not less than 0.5%.
- Duct access panels shall be provided as per AS1668.1 (i.e., at each change in direction and not exceeding 3 m spacing).

Provide a stainless steel engraved ON/OFF control fan switch with speed control and neon indicator mounted at 1100 AFFL.

9.3 Standard AS1668.2 Kitchen Hood

The hood shall be provided with a grease gutter around the base of the hood, not less than 50 mm wide or more than 25 mm deep, fitted with a 30 mm brass cleanout plug (minimum 2 No. in diagonally opposite corners) that does not impede the flow of grease in the gutter.

The internal edge of the grease gutters of the hoods shall extend not less than the overhang distance required in accordance with the cooking process as stipulated in AS1668.2 clause 3.5 beyond the perimeter of the cooking equipment. (i.e., between 150 mm minimum and up to

600mm for type 6 cooking process). Check equipment dimensions prior to commencing manufacture. All dimensions noted on the drawings are nominal only.

Design the hood duct take-off and internal airflow path for low air velocity, pressure drop and noise generation. Ensure the internal airflow path does not cause uneven distribution of air flow through the filters. Noisy hoods will be rejected.

9.3.1 Lights

Kitchen exhaust hood lighting to be provided as indicated on the equipment schedules and/or drawings. Lights shall be installed equally above all cooking equipment. Provide luminaires as nominated below:

Lighting shall be sealed LED panels to meet the code-required lighting levels.

Connect wiring from the luminaires to a junction box. Electrical services will connect the lighting circuit to this junction box.

9.3.2 Kitchen Canopy Grease Removal Filters

The filter media and holding frame shall be constructed of rigid material and deemed non-combustible when tested in accordance with AS1530.1. Each panel shall be clipped into an individual frame fixed to the canopy. Filters shall be sized to achieve a maximum face velocity of 1.8 m/s.

Filters and filter-retaining devices shall be flush mounted, either vertical or sloped at an angle not greater than 30° from vertical.

The filters shall be easily removable for maintenance and cleaning purposes.

Kitchen exhaust filters shall be compliant with UL 1046 [applicable when the exhaust duct within the building exceeds 10 m and an open flame (including gas range etc.) is used.]

9.3.3 Kitchen Canopy Grease Removal Devices

Grease removal devices other than filters shall remove grease from cooking vapours and they shall prevent grease from falling back into food, the cooking appliance or on the floor.

The grease removal devices shall be automatically or manually cleaned, including all internal surfaces of device housing, and device shall be demonstrated to be effective at grease capture and removal with efficiency equal or greater than grease removal filters.

10. AIR DISTRIBUTION SYSTEM

10.1 General

All sheet metal air distribution systems required for the works and including all necessary control dampers, air diffusers, registers, grilles, insulation, and painting, all as specified in the relevant clauses of this specification and as shown on the drawings, shall be supplied as part of the Mechanical Systems.

10.2 Standards

The following standards are applicable to this section:

AS 4254	Ductwork for air-handling systems in buildings
AS 4508	Thermal resistance of insulation for ductwork used in building air conditioning
AS 1530	Test for Early Hazard Properties of Materials

10.3 Rectangular Ductwork Construction

Ductwork shall be fabricated from first quality patent-flattened galvanised sheet steel.

Ductwork construction shall be based upon SMACNA Standards and AS 4254.

Rectangular ductwork shall be made from full size sheets and any ductwork comprising patched or pieced-up sheets shall not be permitted. All panels will be cross broken for rigidity, or alternatively, shall be of two (2) sizes heavier than those specified. Ductwork shall be free from any sign of waves or buckles and all sheets shall be machine bent to ensure neat and accurate fabrication.

Longitudinal seams shall be Pittsburgh lockseam type for 0.9mm and lighter and knock-up seam for heavier gauges. Intermediate joints shall be made by drive cleats, properly sealed to approval, or of approved patented, well known flanged and bolted manufacture.

Proprietary flange systems used for duct jointing shall incorporate neoprene gaskets, or equal approved, securely glued to one flange to prevent incorrect alignment. Supplementary sealing using suitable jointing compounds shall be applied as necessary.

Ductwork shall be supported to approval at 1.5 to 3.5 metre intervals according to size and construction of the ducts. Supports, generally, shall comprise galvanised angle slings under the duct with galvanised suspension rods carried from brackets secured to the roof structure. Such rods will be screwed to facilitate proper levelling of the ducting. Vertical ducts shall be supported by galvanised angle cantilever bracket.

Horizontal dimensions of rectangular ducts are shown first in plan views. All dimensions shown are clear internal sizes.

10.4 Sealing

Heating or cooling ductwork and fittings must be sealed against air loss by closing all openings in the surface, joints, and seams of ductwork with adhesives, mastics, sealants, or gaskets in accordance with the duct sealing requirements of AS 4254 for the static pressure in the system.

10.5 Volume Control Dampers

Volume control regulating dampers shall be fitted in main duct branches where shown on the drawing and as necessary for system balancing and in return air, exhaust air and fresh air inlet ducts or openings where balancing is required.

Dampers shall be constructed of extruded aluminium and linked to operate in an opposed blade arrangement. Damper blades shall be rigidly fixed to spindles of not less than 10mm with suitable operating devices. Stops shall be provided at fully opened and closed positions and provided with a locking quadrant for fixing dampers in any desired position. Damper blade direction shall be shown by indicating devices and open and closed positions shall be neatly labelled.

Dampers shall be of proprietary galvanised steel or aluminium construction with opposed blades of minimum 150mm width, complete with edge and blade seals, linkages, etc.

Supply and return air dampers shall be externally wrapped with 75mm thick insulation with impermeate foil vapour barrier and, where exposed, weatherproofed. Place return air and outdoor air dampers within mixing plenums where possible.

10.6 Modulating Dampers

10.6.1 Construction

Modulating dampers shall be aluminium and shall be constructed in accordance with requirements specified below.

10.6.2 Sealing

Motorised dampers shall be fitted with blade edge and blade end seals. Seals shall be of a standard such that total air leakage does not exceed 30 l/s per m² of damper area when closed against static pressures of 250Pa.

Blade seals shall be mechanically fastened in position. Seals which are glued on will not be accepted.

10.6.3 Shafts

Shafts shall be securely fixed to the centre of each blade. Driven shafts shall run the full length of the blade and extend to provide suitable attachment to the drive motor.

10.6.4 Bearings

Blade shaft bearings shall provide free rotation of shafts and shall be of brass construction. Bearings in fire smoke control dampers shall be sintered bronze, ball, or roller type.

10.6.5 Linkages

Dampers shall be interconnected to give the required parallel or contra rotating action.

10.6.6 Motors

Motors shall be rigidly mounted to prevent flexing and shall be linked to the driven blade shaft extension using suitable linkages. Dampers required to control areas greater than 1.6m² shall be divided into multiple sections, each of 1.6m² maximum area with an individual drive motor for each section.

10.7 Flexible Connections

Approved flexible connections shall be provided between the inlets and outlets of all air handling equipment and adjacent ductwork and shall be truly aligned to ensure smooth air flow.

Flexible connections shall be readily accessible for maintenance and removal and shall be positively airtight under full fan pressure.

In locations where there is movement in the ground, flexible connections shall be of adequate length and movability to allow for any plant movement due to local ground conditions.

10.8 Circular Flexible Ductwork

All flexible ductworks shall comply with AS 4254 and AS 4508. Flexible ductwork shall be suitable for the duties and pressures as required for the system.

Duct core shall be constructed with an aluminium/polyester film laminate incorporating a coated mechanical spring wire helix and solvent based adhesive.

The polyester insulation blanket surrounding the core shall be encapsulated in a UV resistant polyethylene sleeve or aluminium / polyester laminated sleeve for the fire rated version.

For ventilation and relief air applications the core shall be constructed with an aluminium / polyester film laminate incorporating a coated mechanical spring wire helix and solvent based adhesive.

All bends made in flexible ducting shall be formed to manufacturer's recommendations and shall have a centre-line radius of not less than 1.5 times the diameter of the ducts. Bends shall remain at full diameter throughout the length of the bend and all bends deformed or damaged in any way during installation shall be replaced with new bends.

All fittings such as tee-pieces, Y-pieces and reducers shall be of galvanised metal exactly as specified for circular sheet metal ductwork.

All connections to fittings and terminal boxes etc. shall be of the spigotted slip joint type.

Duct connections shall be taped and provided with bands as follows.

All joints shall be fixed with proprietary line duct bands. These shall be continuous type metal or PVC bands similar to "Jubilee" hose clips or approved equivalent may be used. Self-tapping metal screws and rivets shall not be used.

All flexible ducts shall be adequately supported with hanger straps and half saddles of "gutterguard" located at 1500 maximum centres, to prevent undue sagging and strictly in accordance with the requirements of this section of the specification.

Circular flexible acoustic ductwork shall be of the metal helix type incorporating perforated acoustic lining and reinforced external vapour barrier.

10.8.1 Fire Performance

The ductwork shall have the following minimum indices when tested in accordance with AS 1530.

Ignitability	-	0
Spread of Flame	-	0
Heat Evolved	-	0
Smoke Developed	-	3

10.8.2 Sealing

Flexible ductwork at an operating static pressure of less than 500 Pa must be sealed against air loss with a draw band in conjunction with a sealant or adhesive tape.

10.8.3 Thermal Resistivity

Flexible ductwork used for supply and return air applications shall be of the patent pre-insulated type, compliant with the NCC Part J6D6 (1) (b) (i). i.e.: must achieve a minimum R-value of 1.0.

10.8.4 Flexible Duct Length

Flexible ductwork length in any duct run shall be limited to 6m in compliance with the requirements of the NCC Part J6D5 (3) (b).

10.9 Circular & Oval Ductwork

Solid sections of circular and oval ductwork shall be fabricated using a minimum of 0.8mm thick heavy ribbed (spiral joint only) Duraduct or approved equivalent.

Duct lengths shall be as long as possible with a minimum of site joints.

Site joints shall be reinforced with galvanised mild steel reinforced spigot and shall be arranged so that the spiral joint matches in a continuous method. The joint shall be sealed with an approved mastic sealer which shall be wiped clean after assembly to give a smooth joint. Badly made and open joints will be rejected.

Any dents blemished or exposed fixings will be rejected if judged to be unsightly on completion of painting.

The internal section of the ducts shall be reinforced to prevent sagging or distortion of the duct between hangers. Sagging ductwork shall be rejected.

10.10 Duct Test Holes

Provide duct test holes in ductwork on discharge and intake side of fans and fan coil units. The test holes shall be such as to enable pitot tube and/or velometer readings to be taken using 38mm diameter holes plugged with rubber grommets shall be provided for this purpose.

10.11 Access Panels

10.11.1 Kitchen Exhaust

Provide clean out access panels 400 x 300 every 2m to enable easy access for cleaning kitchen exhaust ductwork and dishwasher exhaust ductwork.

10.12 Insulation

10.12.1 General

Ductwork must be insulated when used in a system that cools or heats a building.

Insulation must:

- be protected against the effects of weather and sunlight
- abut adjoining insulation to form a continuous barrier
- be installed so that it maintains its position and thickness, other than at flanges and supports
- be able to withstand the temperatures within the piping, vessel, heat exchanger or tank
- have an ozone depleting potential (ODP) of zero

10.12.2 Duct Liners and Insulation

The following shall apply to rigid ducts:

- a) Duct liners shall comply with the following requirements:
 - I. Bulk insulation shall have a smoke developed index not greater than "3" and spread of flame index not greater than "0" when independently tested in accordance with AS1530.3
 - II. Duct liners assembled on a duct system, i.e., the assembled final duct, shall have a smoke developed index not greater than "3" and spread of flame index not greater than "0" when tested in accordance with AS1530.3
 - III. Duct liners assembled on a duct system, i.e., the assembled final product, shall pass the UL 181 burning test

10.12.3 Thermal Resistivity

Ductwork must be insulated in accordance with the NCC Part J6D6, complying with AS4859.1 in accordance with the following R Values:

Table J6D6: Ductwork and fittings - Minimum Material R Value

Location of Ductwork and Fittings	Climate Zone	
	1, 2, 3, 4, 5, 6 and 7	8
Within a conditioned space	1.2	2.0
Where exposed to direct sunlight	3.0	3.0
All other locations	2.0	3.0

Thermal resistivity of insulation material must be tested in accordance with AS4859.1. Documentation indicating as such must be provided on request.

10.12.4 Ductwork sealing

Ductwork in an air-conditioning system with a capacity of 3000L/s or greater, not located within the only or last room served by the system, must be sealed against air loss in accordance with the duct sealing requirements of AS4254.1 and AS4252.2 for the static pressure in the system.

10.12.5 Vapour barrier

Insulation on ductwork conveying cold air must be protected by a vapour barrier on the outside of the insulation. Where the vapour barrier is a membrane, adjoining sheets of the membrane must overlap by 50 mm and bonded or the sheets are to be taped together.

10.12.6 Insulation at Flexible Connections

Flexible connections are to be insulated in accordance with *10.12.3 Thermal Resistivity*.

Where connections are exposed to the weather, protect the connection with a weatherproof sheet metal cover.

10.12.7 Insulation of Registers

Provide insulation to the rear exposed section of supply air registers as follows:

- Registers within 4m of a doorway to a non-air-conditioned space
- Registers in a roof space or other non-air-conditioned space

10.13 Air Terminal Equipment

10.13.1 General

All air diffusers, registers, and grilles necessary for the installation shall be supplied generally as noted on the drawings and specified. The drawings show provisional locations but, in each instance, outlets shall be installed in accordance with Architectural final detail drawings and reflected ceiling plans.

Damper assemblies and internal surface of outlets and take-off ductwork, where visible through outlet faces, shall be painted matt black.

Diffusers shall be finished powder coat, colour to be nominated later (allow cost of non-standard powder coat colour)

10.13.2 Supply Air Diffusers

10.13.2.1 Swirl Diffusers

Swirl diffusers shall be Holyoake CFP swirl diffusers or approved equiva, flush mounted in the ceiling.

The flat face of the diffuser shall 600 x 600 and made of powder coated galvanised steel.

10.13.3 Return/Exhaust Air Registers and Grilles

10.13.3.1 Return/Exhaust Air Registers and Grilles - Ceiling Mounted

Return/exhaust air registers and/or grilles shall be constructed with aluminium lattice “egg-crate” cores of 13mm x 13mm grid and frames to suit each location and shall have hinged cores only if nominated on the drawings. Registers only shall incorporate opposed blade dampers.

Ceiling mounted return air registers and grilles shall be Holyoake model EC-125 or approved equal.

10.13.3.2 Return/Exhaust Air Registers and Grilles - Wall Mounted

Wall mounted return/exhaust air registers and/or grilles shall be constructed with heavy duty aluminium half chevron blades at 25mm centres, suitably stiffened and mounted in aluminium frames.

Registers and grilles shall be of Holyoake model RLW or approved equal.

10.13.4 Door Grilles

Door mounted relief grilles shall be constructed from aluminium extrusions with full chevron blades and shall incorporate back frames with concealed fixing and be screw fixed with countersunk screws.

Grilles shall be of Holyoake model DG-52 or approved equal.

OR

Grilles shall be of Holyoake model DG-17 or approved equal.

10.13.5 Weather Louvres

Weather louvres shall be constructed from aluminium extrusions. Blades shall be on approx. 50mm centres. Louvres shall be finished powder-coat, colour to be nominated at a later date (allow cost of non-standard powder coat colour).

Louvres shall be of “Holyoake” model OHL-45 or approved equal.

10.13.6 Plenum Heads to Outlets, Exhaust Grilles, and R.A. Grilles

Provide cushion heads to all ceiling grilles. Sizes of plenum heads shall be to suit neck, inlet spigot and supply air quantity, and shall be internally insulated with rockwool faced with black faced insulation.

Boxes shall be constructed from galvanised steel in accordance with 'Steel Ductwork' requirements and shall be internally insulated in accordance with 'Insulation of Steel Ductwork' requirements.

For inaccessible ceilings, or where indicated on the drawings, provide a balancing damper retained in friction mounts on the spigot inlet to the box. The balancing damper shall be adjustable through the diffuser face.

Boxes shall be supported from the structure above by means of galvanised vertical supporting straps to each corner. Where necessary to ensure vertical support, install galvanised angle bridging supports between purlins or other structural elements.

10.13.7 Cushion Head Boxes (Lightweight)

Supply and install lightweight insulated cushion head boxes for air diffusers where shown on the drawings. Boxes shall be constructed from sandwich panel of aluminium foil/polyurethane foam/aluminium foil construction (foil faces shall be polyester lacquered), of "P3 Ductall" or equal manufacture. Boxes shall be provided with approved plastic reinforcing extrusions on the bottom edge and suitable reinforcement at duct spigot connections. Dampers shall be at main duct branch spigots. Boxes shall be fastened and sealed to the diffuser or register by approved means, to prevent air leakage or dislodgement of the box (submit details for approval). Boxes shall be painted internally with a matt black finish.

11. ELECTRICAL

11.1 General

The Electrical Installation associated with the Mechanical Services shall be supplied and installed to Standards Australia Wiring Rules AS3000, and the following specified requirements.

Provide compliant RCD protection with AS3000 where required.

Each item of equipment shall be able to be electrically isolated at the item of equipment.

11.2 Power Supplies

Electrical Services will provide a suitable three (3) phase neutral and earth sub-main power supply adjacent the mechanical services switchboards (MSSB-1 & MSSB-2).

All wiring from the MSSB to mechanical equipment in the field shall be by Mechanical Services.

Any switch that is capable of isolating power in fire mode associated with fans required to operate during a fire shall be lockable and shall be fitted with a prominent warning label as follows:

11.2.1 Earthing

All Mechanical services are to be earthed in accordance with AS3000.

Examples being:

- Ductwork including across flexible connections
- Pipework including across pump bellows
- Equipment bases / frames
- Switchboards

11.3 Air Conditioning Units

Provide for all isolators, controls, wiring and connections to all equipment including all wiring for controls.

All wiring to thermostats, etc. shall be concealed throughout by running in the ceiling space and walls.

All air conditioning plants shall automatically reset in the event of power failure after the appropriate interval recommended by the manufacturer.

11.4 Fire Shutdown

Provide a fire trip relay in the switchboard to shut down the air conditioning in the event of a fire alarm. Liaise with the Fire Services for the relay coil voltage required. The Fire Services will provide wiring and final connection to the fire trip relay.

Mechanical Services shall wire from the isolator to the above equipment to be shut down on receipt of a fire alarm signal. Liaise with the Fire Services contractor to obtain details of the signal to be provided.

11.5 Method of Wiring

All mechanical services equipment shall be wired in accordance with the requirements of AS3000.

The entire installation shall be satisfactorily balanced to the approval of the Supply Authority.

All wiring shall be installed on the loop-in system with connections made only at equipment. The use of junction boxes shall not be permitted.

Cables shall be supported over their entire route length on trays or grouped together and supported at regular intervals to minimise sag.

Separate earth wires shall be run for all circuits and shall originate at the switchboards and run in the same conduit or within the TPS cable of the circuit.

Wiring to equipment shall be installed in conduits, cable trays or wiring ducts within accessible areas, and neatly supported to the *Wiring Code* for inaccessible areas.

All wiring shall be installed concealed in concrete slabs, walls, partitions, and the ceiling space, unless otherwise specified.

The drawing-in of cables into conduits shall be performed by means of a draw wire, and the total number of cables shall be drawn in at one time in each section of the pipe. The number of cables within conduit shall be 75% of that allowed under AS3000. Should more than one (1) circuit be installed in any conduit, cable sizes shall be increased to allow for derating as per AS3008.1.

Cable shall be installed to avoid derating in accordance with AS3008.1.

Chasing of face brick walls is not permitted except where 110mm brick walls will be rendered or plastered.

Wiring within wall cavities and ceiling spaces shall be in TPS conductors specified herein.

Wiring to equipment mounted on brick walls shall be installed in conduits.

Where the locations of equipment are critical and do not coincide with stud locations, supply, and install suitable brackets or noggings, etc., as required to ensure equipment is mounted securely in the correct locations.

Where conduits or TPS cables are installed in external cavity brick walls, they shall be fixed to the internal skin. Penetrations through the damp course shall be at the highest point.

Liaise to ensure that the installation of conduits and brickwork are co-ordinated.

All wiring runs shall be checked before installation commences and where it is considered that conduits or wiring cannot be concealed, notify, and obtain approval before installation commences.

11.6 Cables

11.6.1 General

All control and power sub-circuits for equipment shall be wired using stranded copper conductors of approved manufacture, grade V75 PVC insulated and 250V or 660V grade as applicable. All cables shall be from new stock.

Building wires shall be black for neutral and coloured for active and control wires. TPS cables shall have the earth continuity conductor incorporated in the same sheathing as the circuit wiring.

11.7 Conduits

All conduits and ducts shall comply with the requirements of the relevant Australian Standard. No conduit less than 20mm shall be used.

All conduits shall be high impact PVC conduit to AS 2053.

Conduits shall be laid in straight and symmetrical lines with easy sets or bends. Bends or sets must be made without altering the section of the conduit. Where it is not practicable to set the conduit, spring bends may be used. PVC conduit shall be jointed using the manufacturer's recommended solvent and following closely his printed instructions.

Conduits installed in screeds and chases shall be secured at intervals not exceeding 1800mm with approved fixings. Conduit stub-ups through concrete surfaces and foundations shall be adequately braced to prevent shifting during pouring and shall be adequately plugged to prevent ingress of dirt. Surface mounted conduits shall be fastened at intervals not exceeding 900mm using approved saddles.

12. CONTROLS

12.1 General

The air-conditioning, ventilation and other mechanical services plant shall be provided with a fully automatic control system to control the plant as indicated herein.

12.2 Air Conditioning Controls

- Manufacturers permanently wired LCD controls stations are to be used.
- The sensors shall be located where shown on the drawing. The remote control shall have a calibrated dial, readily accessible for adjustment.
- Samples of control units shall be submitted for approval prior to installation.
- Provide multichannel or single channel time clocks to control each air conditioning unit. Each channel to be individually programmable. Provide 7-day clocks with 24-hour battery backup. Provide a holiday isolate and a time clock bypass switch at the units control station.

12.3 Air Conditioning Units (Packaged Systems)

Provide air conditioning unit manufacturer's wall mounted wired controller for each ducted indoor unit. All controllers shall be provided with a liquid crystal display of the following minimum features:

- On / off control with green run light
- Temperature setting
- Fan speed
- Operation mode switching – cool / dry / auto / fan / heat.
- Weekly time schedule
- Auto stop timer.
- Temperature sensor at the controller or where indicated.
- CO₂ Control: Provide a CO₂ sensor in the return air, where nominated in the Schedule, to modulate outdoor air dampers or speed control outdoor air supply fans, to maintain CO₂ level between 600 and 900 ppm (adjustable via the BMS) using proportional control. Outdoor airflow is not to drop below minimum figures in the mechanical AC schedules, or to be determined in consultation with the Managing Contractor
- Economy cycle: Economy cycle damper(s) shall be opened automatically in the event of outside temperature conditions from 20°CDB to 15°CDB and relative humidity between 80% and 20%. Temperature and relative humidity limits shall be adjustable.

12.4 Air Conditioning Control Station

Supply and install manufacturers control station(s) as indicated on the drawings.

12.5 Kitchen Exhaust Fan

In the event of a fire alarm kitchen fan shall start and supply fan shall stop. Fans speeds shall be electronically interlocked. Provide timers to introduce a delay between speed changes both increase and decrease.

12.5.1 Lights in Canopies

Luminaires shall be provided with a three-pin plug top and flex and plugged into a plug base specified in the electrical services.

Circuit wiring including switching of luminaires will form part of the electrical services installation.

13. MECHANICAL SERVICES SWITCHBOARD

13.1 General

The switchboard shall be manufactured from 316 marine grade stainless steel, or marine grade aluminium.

Current BAC MSSB suppliers:

- NHP Switchboards
- Schneider Electric
- Actenium (formerly J&P Richardson)
- Trivantage Manufacturing
- Switchboard Innovations
- Sunline
- GB Electrics
- LAI
- Reid

Switchboard manufacturer requires BAC approval.

The switchboard must have NHP Grizz-Bar or Schneider Isobar chassis.

13.2 Installation

Provide for the supply, installation, final connection, and commissioning of the mechanical services switchboards where shown on the drawing.

Particular attention shall be made in the design of the switchboard to ensure satisfactory handling into position on site during construction and that the switchboard is designed to suit installation requirements.

13.3 Standards and Rules

The design of the switchboard shall be in accordance with the relevant Australian Standards, the additional requirements of this specification, and shall meet the necessary requirements of the Local Supply Authority having jurisdiction over the electrical installation.

Provide compliant RCD protection with AS3000 where required.

All MSSBs shall:

- be dead front, metal clad, moulded case (MCCB) and miniature circuit breaker (MCB) type.
- be provided with testing and verification documentation to AS/NZS 61439 / AS3439.
- have Form of Separation (Form 1 – 4) to AS/NZS 61439/AS3439 as required by AS/NZS 3000.
- have a degree of protection of at least IP42 to 60529 were installed indoors, increasing to a minimum of IP56 where in a path of travel to an exit.
- be current rated for maximum in-service operating conditions.
- be fault current rated to withstand the maximum prospective short circuit current for 1 second.
- have 25% spare rated capacity and minimum 25% spare equipment/MCB pole capacity.

- be of steel construction with removable escutcheon plates, lockable doors, cable entry provisions and mounting provisions finished with factory applied corrosion protecting paint finish – white interior and escutcheon, grey or orange exterior.
- have corrosion resistant components of non-ferrous metals or chromium plated steel.
- have solid copper busbars, colour coded and insulated.
- have fault make, load break isolators on MSB.
- be fully labelled (dual colour engraved plastic sandwich type) to indicate equipment function, rating, circuit number, etc.
- be complete with all necessary timeclocks, contactors, relays, control switches, etc. for systems functions as shown (including emergency lighting test switching to AS/NZS 2293.1).
- have type written circuit schedules installed behind hard plastic transparent covers.
- have removable links for current transformers.
- have combined MCB/RCD's where required for AS/NZS 3000 compliance.
- Have all cables installed through gland plates with compression glands.

Additional requirements for externally mounted MSSBs:

- IP56 minimum.
- Bottom entry only, unless under a roof with 45-degree protection again wind driven rain.
- Sloped rain hood with gutter, extending 50mm beyond the extents of the enclosure.
- Be of 2.0mm marine grade stainless steel or 3.0mm powder coated marine grade aluminium construction.
- Be fitted with anti-condensation heaters and thermostats in each section which contains switchgear or BMS equipment.

13.4 Switchboard Drawings

Before construction of the switchboard is commenced, electronic PDF drawings shall be submitted for approval. The drawings shall clearly indicate the following information:

- Manufacturer's name and type of any standard equipment.
- General arrangement of equipment.
- Full details of cabinet construction and dimensions.
- Sheet metal details and finishing details.
- Front elevation and vertical section through each compartment.
- The size and wording of labels.
- Method of supporting busbars and equipment.
- Description of all materials to be used.
- Busbar dimensions and ratings.
- Clearances between live parts, and live parts and earth.
- Internal wiring sizes and ratings.
- Compliance with AS/NZS 3439.1 or AS/NZS 61439.
- IP rating to AS60529.

- Maximum fault withstand ability.

Any approval of drawings given shall in no way relieve the manufacturer of his obligations under other sections of the specification.

A copy of the "as installed" Schematic Diagram for the switchboard shall be provided. Diagrams shall be suitably framed and mounted adjacent to the switchboard.

13.5 Earthing

All metal parts of the switchboard and associated equipment shall be bonded and connected to a main earth bar. All earthing conductors shall be made strictly in accordance with the requirements of the Supply Authority.

13.6 Labelling

All equipment shall be clearly labelled in accordance with regulations and to indicate its function by means of engraved "Traffolyte" labels.

Main switches shall be "red-on-white"; other shall be "black-on-white".
Letters shall be not less than 10mm in height.

13.7 Construction

The switchboard shall be metal clad, totally enclosed, lockable, dustproof/weatherproof type of construction.

Switchboard shall be manufactured from 1.6mm minimum C.R.C.Q. sheet steel pressed or rolled to shape. All joints shall be neatly welded and finished flush with the adjacent surfaces by grinding and/or machining. Joints shall not be located on a corner and all bare edges shall be lipped.

The cabinet door furniture should be:

- The following equipment provides 3-point locking. The equipment is all Selectrix and the applicable part no.
- Selectrix Stainless Steel Pad lockable Swing Handle 1107-SS02-316.
- Selectrix cam 1000-1588-SS.
- Rod Adapt SS-2 required.
- Rod – 1200RSS-2 required.

No lift-off hinges will be acceptable. All swing doors to have compliant minimum clearances in open position.

Structural members and bracing where necessary shall be welded or bolted to the frame.

All cut-outs for panel mounted accessories shall be neatly made and additional supports shall be provided where necessary to avoid panel distortion.

Escutcheons to be white and external surfaces grey or stainless steel finish.

13.8 Finishing of Metalwork

The switchboard shall be etched primed followed by one (1) coat of undercoat and finished externally with two (2) coats of high gloss baked enamel and internally with one (1) coat of high gloss baked enamel, white.

Stainless steel enclosures shall be brushed/linished finish.

13.9 Main Isolator

Main isolator shall be flush panel mounted type rated to suit equipment.

The main isolator shall be rated appropriately for the required upstream protection.

13.10 Contactors

Contractors shall:

- Contactors shall be AC3 rated to suit motor sizes.
- Comply with AS60947.1 and AS60947.4.1.
- Be Siemens, Nilsen or approved equal manufacture.

13.11 Circuit Breakers

Circuit breakers shall be used throughout the installation unless shown otherwise. Circuit breakers shall be of one manufacture throughout. Liaise with Electrical Services to ensure both systems utilize the same brand and model of circuit breakers to ensure uniformity throughout.

Circuit breakers shall be rated for the calculated prospective short circuit fault level.

Mechanical Services shall ensure that protective devices are selected and adjusted correctly to ensure reliability of supply. A fault anywhere in the mechanical services installation shall not interrupt supply to any other part of the installation (i.e., full discrimination).

Circuit breakers shall have visual indication of the "tripped" position independent of the "off" position.

Circuit breakers shall be mounted on a standard mounting chassis of approved manufacture securely fastened to a channel fixed to the back of the switchboard enclosure.

Construction of the circuit breaker assembly shall be such that a double or triple pole circuit breaker can be replaced by three (3) single pole circuit breakers and vice versa, without modification to the busbar assembly.

Cabling zones 125mm wide (minimum) shall be provided between the rows of circuit breakers. Additionally, a minimum distance of 125mm for cabling shall be provided between circuit breakers and the top, bottom, and sides of the switchboard cabinet. Circuit breakers shall be numbered with push-in circular dish type numbers in the appropriate phase colours fixed to the escutcheon plate. Circuit breakers shall be further numbered within the switchboard to correspond to the numbers on the escutcheon.

13.12 Busbar and Wiring

Three (3) phase copper busbar assemblies shall be provided to the ratings as required by the connected loads plus minimum 30% spare capacity.

Busbars shall be mounted independently of the circuit breakers and shall be insulated with 660V grade insulation of appropriate phase colours for their entire length - "A" Phase - RED, "B" Phase - WHITE, "C" Phase - BLUE, Neutral - BLACK.

All busbars and current carrying parts shall be of a copper section and shall be capable of carrying the rated current continuously without the temperature rise of any part exceeding 40 °C.

Neutral busbar shall be rated equal to the phase busbar. The minimum size busbar shall be 13mm x 3mm copper.

Ensure that the minimum busbar sizes and support spacings withstand fault conditions as required for safe operation.

Neutral and earth bars shall be provided within each switchboard and the number of connections shall correspond to the number of initial and future circuits. The connections shall be numbered to correspond with the circuit numbers.

Adequate provisions shall be made for connection of mains and sub-main cables by means of cable lugs.

All connections shall be bolted or clamped in an approved manner and fixed with flat washers and locknuts.

13.13 Switchboard Facia

The following shall be provided on each switchboard:

- Auto/Off/Manual switches for fans.
- Fault and run lights.
- Test push buttons.

13.14 Pilot Lights/Fault and Run Lights

Provide on the panel fascia (or behind the door in the case of a weatherproof external panel) fault and run lights for each of the mechanical drives not provided already with such facilities on their own on-board panels.

Pilot lights shall be of the dual LED type installed to indicate the following functions for motors and associated equipment. Generally, lamp colours shall be as follows:

Green	-	Run
Red	-	Fault
Amber	-	Power on
Blue	-	Fire trip

Fault indication shall be obtained from each unit's general fault circuit.

Pilot lights for fan motors or other associated equipment shall be grouped logically on the fascia panel clearly labelled to denote their functions. Pilot lights shall be connected via auxiliary contacts to the load side of the control fuses and not parallel connected with the starter or relay operating coil.

Pilot and indicator lamp circuits shall be designed to ensure long lamp life consistent with clear visibility in the surrounding level of illumination but shall be clearly "off" when de-energised.

A lamp testing circuit shall be provided incorporating a common push button and isolating diodes. Diodes shall be rated for 1 amp continuous with a peak inverse voltage of 400V.

13.15 S.S.O. and Light

Provide a double switched socket outlet on mounting panel at low level.

Provide LED bar light at high level in each module, operated automatically by door switches.

13.16 Test Switches

Provide for testing and operational purposes, test switches for all nominated major plant as nominated by the superintendent's representatives. Switches to be Kraus and Naimer CG4 rotary on / off / auto and provided for:

- Ventilation fans

13.17 Phase Failure Protection

Provide reliable phase failure / phase rotation protection for all 3-phase equipment.

13.18 Spare Space

Each switchboard shall be sized and arranged to allow for future expansion. Spare space for a minimum 25% increase in components in all sections of the switchboard shall be incorporated.

13.19 Segregation

Where essential and non-essential sections are specified for switchboards, proper segregation shall be incorporated between sections.

13.20 Maintenance

13.20.1 RCD Testing

For all switchboards and distribution boards, carry out testing of all RCDs and submit report one month prior to the end of the defect's liability period.

Carry out tests for the operating time of each RCD in accordance with AS/NZS 3760 and record the test result in the report.

Replace (at no cost) any RCDs which do not meet the requirements of AS/NZS 3760 during the defect's liability period.

13.20.2 Thermographic Test Report

Carry out thermographic scans and submit report 1 month after practical completion, and again one month prior to the end of the defect's liability period, for all switchboards.

Thermographic scanning shall be conducted under full load conditions. Coordinate with the building operator to ascertain when this is and carry out the scans at this time.

The load currents on each phase for each outgoing functional unit or conductor shall be measured at the time of the scan and logged in the report.

Carry out all corrective actions recommended in the report at no cost.

14. NOISE AND VIBRATION

14.1.1 General

All air handling and water systems, pipework, ductwork, grilles, and all associated mechanical equipment shall be selected and installed for quiet and vibration free operation in compliance with the respective sections of this specification and within the noise and vibration criteria as specified for the individual items of equipment, acoustic linings, and vibration isolation.

Any item of plant or equipment found to be causing undue noise and vibration within the occupied spaces and which exceeds specified sound power or vibration levels shall be replaced as directed. Similarly, noise attenuation and vibration isolation equipment which does not perform as specified shall be replaced or rectified.

The minimum anti-vibration mountings acceptable are:

Equipment	Mounting
Large condensing units	Double waffle pads with galvanised shim dividing plate, and cyclone bolts
Ducted fan coil units	Rubber in shear mounts
In-Line fans	Rubber in shear hangers or mounts

Ensure all noise levels at site boundaries are compliant with local government regulations.

14.1.2 Extent of Work

All mechanical services shall comply with the following requirements to ensure that noise and vibration from the plant installed under this contract is reduced to a minimum.

- All rotating machinery shall be accurately balanced both statically and dynamically. Motor impellor rotors shall be balanced to international standard ISO 1940 – “Balance Quality of Rotating Rigid bodies” and shall be free of vibration at all operating speeds and during starting and stopping.
- Centrifugal and reciprocating rotating equipment shall be mounted on vibration absorbing mountings.
- All connections to rotation machinery or assemblies containing machinery shall be rendered flexible by vibration hangers supporting the ducting and piping systems, flexible connections between ductwork and fans, and in critical installations with flexible hoses between pipes and pumps. If flexible hoses are not installed, adequate provision shall be made to take up vibration in bends and pipe runs.
- Acoustic lining shall be provided to critical sections of ducts and air handling units unless otherwise specified.
- Acoustic seals shall be provided where all pipes, ducts and conduits penetrate plantroom or acoustic walls.

- All equipment shall not have sound power levels greater than those specified.
- All terminal boxes, air outlets and similar equipment shall be selected within the specified operating noise ratings and shall be lined with acoustic material and/or provided with cushion heads as appropriate.
- All duct interiors, air outlets, dampers and similar air distribution equipment shall be free from raw edges of metal, sharp screw ends or pop rivets projecting into the air stream.
- All static equipment in major plantrooms shall be mounted on 18mm Mason Super W isolation pads, loaded to between 250 and 400kPa.
- Duct connections to vibrating mechanical plant shall be isolated by flexible PVC connections not less than 150mm long when fully stretched out.
- Generally, vibration isolation shall be arranged so that all items of equipment and its associated connecting systems shall be isolated from the building so that there is no dependence on flexible connectors as the main means of vibration isolation.

14.1.3 Isolation Mount and Hanger Minimum Performance

To control structure borne noise and to reduce possible vibration induced structural damage the following minimum levels of vibration isolation shall be used.

Note: All neoprene pads are to be loaded between 250 and 400kPa.

14.1.4 Vibration Levels

Vibration isolation systems shall be designed to achieve the following maximum structural vibration levels.

<u>Frequency</u>	<u>10 to 100Hz</u>	<u>100 to 1000Hz</u>
Acceleration levels (dB re 1g)	70	60

Vibration levels from equipment will be considered acceptable if their acceleration levels in the building structure does not exceed the building vibration z axis curve for acceleration as given in AS 2670.2 "Evaluation of human exposure to whole-body vibration" figure 2a.

14.1.5 Balancing Criteria

Equipment shall not exceed the following maximum vibration levels.

<u>Equipment</u>	<u>Maximum Peak to Peak Displacement</u>	
Fans and fan coil units	Under 10r/s	0.100mm
	10 - 16r/s	0.075mm
	17 - 35r/s	0.050mm

Equipment which does not meet the appropriate standard shall be replaced or altered to approval. Typical alterations that may be required would be:

- improving the dynamic balance.
- replacing bearings damaged since initial assembly.
- improved alignment.
- replacement of incorrect vibration isolators.
- increasing the mass of the machine mounting.

The objective in requiring these standards to be met is the attainment of the full potential machine life between overhauls. Substandard equipment will not be accepted.

All motors/shafts/fans shall be statically and dynamically balanced to the International Standard ISO-1040 Balance Quality of Rotating Rigid Bodies and shall be free from vibration at all operating speeds and during starting and stopping cycles.

14.1.6 Mounts - General

All vibration isolation mounts required for reciprocating and rotating equipment, pipework and ductwork as separately specified in the relevant section of the specification shall be as manufactured by Mason or Embelton or their approved alternative manufacturer, from the best quality materials and selected to last the full life cycle of the building.

Neoprene type mounts may be used for all static deflections up to and including 9mm however above that, combined springs and neoprene pads shall be used.

All vibration hangers and methods of application shall be to the manufacturer's recommendation and to approval. Full details shall be shown on construction drawings and submitted for approval prior to installation.

14.1.7 Spring Mounts

Spring mounts shall be a free-standing type selected and positioned to produce uniform deflection for all springs.

All spring mounts shall have a ratio of mean coil diameter to compressed length at the specified static deflection of not less than 0.8. Springs shall have a minimum additional travel to solid equal to 50% of the specified deflection.

Spring mountings shall incorporate levelling screws with locking nuts and shall be used in conjunction with ribbed neoprene pads.

Holding down bolts shall be used.

Spring housings, mounting plates and frames shall be hot dipped galvanised.

Springs shall be selected by the supplier for the duty specified and a check shall be made of the surge frequency of the springs proposed compared with the predominant frequency of the machine to be isolated. The surge frequency shall not be within 30% of the predominant frequency of vibration.

14.1.8 Neoprene Pads

Loading of ribbed neoprene pads shall be within the manufacturers ratings and in any case shall not be less than 250kPa or greater than 400kPa.

14.1.9 Anti-Vibration Hangers

Anti-vibration hangers shall contain a spring and double deflecting neoprene cup in series.

Neoprene cups shall have a static deflection of 9mm. The complete hanger shall have a deflection as specified in Section 13.1.5 above.

The first 3 anti-vibration hangers adjacent to the vibration equipment shall be as previously specified except that they shall be provided with the additional capacity of holding the piping at a fixed elevation during installation and shall have a secondary adjustment to transfer the load to the spring and maintain the same position and scale.

Approved types of vibration hangers are Mason type DNHS or PDNS. Where pipework risers are located within 5m of noise sensitive areas consideration should be given to the installation of resilient risers' support guides and anchors for the pipework. Where possible the riser shafts should be located as far as possible from noise sensitive areas.

14.1.10 Duct Movement and Vibration

All ducts shall be carefully designed and provided with all the necessary anchoring and flexible connectors to prevent damage to either the ducts or the building structure due to expansion and/or contraction of the ducts in building and to obviate the transmission of vibration from any motive or noise generating equipment such as fans, air handling units and terminal boxes.

14.1.11 Noise Measurement

Noise measurements shall be taken where necessary, for all strategic locations to assess compliance with the specification. The locations shall be selected and approved prior to

measurement. Measurements shall include not only dB(A) but also the respective octave band levels.

All on site acoustic tests shall be carried out by a recognised acoustical consultant using approved laboratory calibrated sound level measuring equipment.

Measurements of sound pressure levels shall be taken during conditions of maximum sound pressure levels during conditions of maximum load at a time when the building is practically unoccupied with reference to AS 1045 and AS 1055, AS 1081, AS 1217 and AS 1469.

14.1.12 Penetrations

Where ducts, pipework and electrical cable trays pass through high performance or plantroom walls and slabs, it is necessary to acoustically treat the penetrations to ensure that the acoustic rating of the barrier is maintained. The hole is to be packed firmly with a high-density Rockwool and the outer edges on both sides of the penetration is to be sealed for at least 15mm with a plug of resilient sealant, and a fabricated trim angle of galvanised steel affixed both sides of the penetration.

14.2 Seismic Restraint

Allow for supports suitable for seismic (earthquake) restraint to AS 1170.4 where applicable.

Refer to Clause 3.22.

15. COMMISSIONING OF SYSTEM

15.1 General

On completion, the mechanical services systems shall be set to work and balanced to ensure that they operate in conformity with this specification.

Balance all air flow rates to the figures shown on the drawings or scheduled elsewhere in this specification. Carry out all testing and commissioning in accordance with the NEBB Recommendations and Standards.

Commissioning tolerance for airflows and water flows

- Design flowrate -0% +10%

Balancing of ceiling diffusers and exhaust or return air registers shall be carried out using a calibrated self-compensating hood of approved type.

All instruments used shall have been recently calibrated in a NATA-certified laboratory. All balancing equipment shall be provided.

Results of all balancing tests shall be submitted in tabulated form prior to Practical Completion, and final figures shall be incorporated in the Operation and Maintenance Manuals.

Allow in the tender price for a change of pulleys and belts on all new ducted split air conditioning units, or supply and installation of a variable speed controller on directly driven units, to achieve design air quantities.

The Superintendent shall be given the option of being present at all tests and be given a minimum of seven (7) days' notice of commencement of tests.

15.2 Preliminary Adjustments & Settings

Make adjustments and settings necessary for the satisfactory operation of the installation.

Preliminary adjustments and settings shall be made before the installation is deemed to be practically complete, including:

- Setting of pressure switches, overloads, cut-outs, and other protective devices.
- Setting of all thermostats and other controls.
- Balancing of air systems.
- Balancing of water systems.
- Preparation of all test sheet data.
- Preparation of Refrigeration Technician test sheet data indicating quantity of refrigerant in each circuit in kilograms (kg).

15.3 Demonstrate to the Clients Representative and/or Consulting Engineer

The following shall be demonstrated.

15.3.1 Air Systems

- Air flow rates at grilles (supply air, return air, exhaust grilles, air intake grilles)
- Air Dampers (manual and motorised)

15.3.2 BMS Systems

- Alarms
- Control Routines
- Energy Meters
- Calibration of sensors (temperature, humidity, pressure, CO₂)
- Web access including Intranet / Ethernet

15.3.3 MSSB

- Operation of switches & lights
- Controllers

15.4 Testing and Commissioning Plan

The Contractor shall produce a testing and commissioning plan for the mechanical services. The plan shall be submitted to the Builder / Project manager / Consulting Engineer for review eight weeks prior to testing and commissioning commencing.

The plan shall provide the following:

- A clear understanding of the procedures and reporting of plant and equipment
- The required interaction between mechanical, electrical, hydraulic and controls subcontractors
- The timely co-ordination of subcontractors from the commissioning schedule
- The interaction of the base building works with tenants fit out works

The plan shall include the following:

- Inspection Test Plan (ITP) - a series of documents consisting of pre-commissioning and procedures that provide predetermined sign off sheets and procedures that follow both the manufacturers and CIBSE guidelines. This enables the plant/equipment to be commissioned.
- Inspection Test Reports (ITR's) - the sign-off documents that verify the plant/equipment's data, performance, or distribution (of air and/or water etc.) via formalised report sheets.
- Non-Conformance Reports (NCR's) - a reporting mechanism that identifies non-prescribed occurrences in the ITP and ITR outcomes. It also recommends the corrective actions to the responsible party.
- Close Out Report – summarising the ITP's, ITR's, NCR's and execution of the works.
- Hold points.
- Reference Commissioning standards
- Identify Commissioning personnel.
- Identify Test instruments to be used and calibration certificates.

- Work Method statements:
 - I. Air Balancing
 - II. FCU Commissioning
- including safe working procedures.

15.5 Further Adjustments

During the guarantee and maintenance period, make such further adjustments and settings as operating experience shows to be necessary for the satisfactory operation of the installation. Adjust for summer and for winter.

15.6 Tuition

The services of a competent person shall be provided to instruct the Superintendent's Representative and the Client's Representative in the operation of the system and equipment.

15.7 Completion

The whole of the installation shall be left complete and all systems available for full operation after the necessary tests and commissioning procedures have been carried out.

15.8 Labelling

Provide engraved traffolyte labels fixed to all plant and systems. Lettering shall be a minimum 8mm high. Lettering shall be black on a white background.

15.9 Site Waste

All waste, redundant material etc. shall be removed and the building left clean and available for occupancy.

16. PAINTING AND FINISHES

16.1 General

Painting shall be carried out by capable and skilled tradesmen. Paints shall be applied evenly and strictly in accordance with the specific recommendation of the manufacturer and on correctly prepared surfaces.

16.2 Extent of Work

The work shall include painting of all exposed work, equipment, and machine bases, supports and the following items but excluding normally bright and polished metal components.

- Ductwork exposed to view, in the occupied areas, mounted externally and in plantrooms.
- Air diffusion equipment (pre powder coated).
- Conditioner casings (factory finished).
- Roof cowls (factory finished).
- Machine bases, inertia bases, plinths, and floating slabs.
- All plant and equipment which is not finish painted to approval in the factory.
- Switchboards, control panels, conduits, cable trays and electrical ducts exposed to view.
- Hat sections (colourbond to approval)

16.3 Materials

Dressings and all other painting materials shall be the products of an approved manufacturer and of the best quality in their respective types. All painting materials shall be delivered to site in the original manufacturer's sealed and labelled containers.

All paints shall be such that if applied in strict accordance with specific instructions of that manufacturer the finished surfaces will remain of true colour, will not fade, tarnish, peel or otherwise deteriorate for a minimum of three years.

All sealers, primers, undercoats and finishing paints shall be of the type specified by the manufacturer as appropriate for the application. Sealing compounds shall be approved proprietary fillers.

16.4 Equipment and Metalwork

Equipment supplied in a factory finished condition will be acceptable provided that the finish is of high standard, acceptable in colour, and does not require extensive touching up. Otherwise, factory finished equipment shall be cleaned, wire brushed and painted on site with not less than two coats of approved finishing paint.

All other non-galvanised equipment and metalwork shall be painted with one coat of zinc chromate or similar approved metalwork primer and two coats of full gloss enamel paint of the specified colour.

All galvanised steel surfaces shall be cleaned of all oil and other protective coatings used in manufacture and given one complete coat of approved type etching primer before applying the finishing coats as above.

All ductwork exposed to view shall be painted as for galvanised steel surfaces referred to above with special care taken to ensure a high quality of overall appearance and finish. All damaged ductwork, including seams, shall be made good and all minor dents and damages made good with an approved type of motor body filler, prior to painting.

All pipework including fittings and valves exposed to view, shall be cleaned, and painted externally with one priming coat and two finish coats.

All pipes shall have identification colours painted on in the form of bands of colours in accordance with AS 1345 and lettering to approval.

Alternatively, proprietary self-adhesive labels may be utilised if samples are submitted for approval prior to installation.

16.5 Colour

Colour to be as directed by the Superintendent.

17. MAINTENANCE AND SERVICING

17.1 General

Carry out maintenance and servicing for a period of 12 months from date of Practical Completion to the end of the Maintenance and Defects Liability Period.

During this period, without any additional charge to the Proprietor, renew any part of the installation that does not comply with the specified performance.

Regular "maintenance", as defined in AS/NZS 3666, shall commence from the date of practical completion, and cease after the elapse of the defect's liability.

17.2 Standards

The following standards are applicable to this section:

AS 1470	Health and safety at work - Principles and practices
AS 1657	Fixed platforms, walkways, stairways, and ladders - Design, construction, and installation
AS/NZS 1715	Selection, use and maintenance of respiratory protective equipment
AS/NZS 1716	Respiratory protective devices
AS 1851	Routine service of fire protection systems and equipment
AS/NZS 1892.1	Portable ladders - Metal
AS/NZS 2865	Confined Spaces
AS3666	Air handling and water systems in buildings – microbial control
NZ 4302	Code of practice for the control of hygiene in air and water systems in buildings
DA 19	AIRAH Application manual – HVAC&R Maintenance
HB 32	Control of microbial growth in air-handling and water systems of buildings

17.3 Maintenance Access

Provide easy and safe access for cleaning, inspection and maintenance for all equipment covered by this specification.

17.4 Maintenance Procedures

Maintenance procedures shall ensure the safe and proper operation of all systems and shall be in accordance with current standard requirements of the Building Act, AS3666 and Regulations having jurisdiction, Australian Standards, Local Authority Regulations, and the schedule provided in the Installation Manual.

Items not covered by the above shall be maintained in accordance with DA19.

The tender submission shall include recommendations for any further maintenance and servicing procedures required for the equipment offered and such services shall be included for in the tender price.

17.5 Maintenance Intervals

All maintenance procedures and intervals shall be strictly in accordance with Standards and regulations, and the equipment manufacturer's recommendations, adapted as necessary for the installation duty including conditions such as equipment frequent cycling and dusty environment.

Provide emergency service on a 24 hour call out basis.

The last maintenance visit prior to the end of the Maintenance and Defects Liability Period shall be a major visit for complete service.

Provide a full coil clean during the last maintenance visit.

The Superintendent's Representative shall be advised of the proposed service programme for the last major visit not less than 7 days prior to the date of the proposed last visit so that a representative may be present to inspect the equipment.

17.6 Maintenance Records

At the conclusion of each maintenance visit a check list of items serviced shall be completed, the service book shall be signed, and check list shall be submitted to a responsible person on site. Within 7 days of the date of the service visit, a report shall be forwarded together with a copy of the check list to the approving authority.

Provide a copy of all service and fault reports to the Superintendent's representative.

Retention moneys or bank guarantee will not be released until all correctly dated and completed reports have been received by the client at the appropriate times and approved. Should these reports not be completed within the Maintenance and Defects Liability Period, then that period, including all equipment warranties will be extended until the reports have been completed.

17.7 Rectification of Defects

Promptly rectify all defects. Retention moneys or Bank Guarantees will not be released until all outstanding defects notified during the Maintenance and Defects Liability Period have been rectified.

18. TENDER FORM A – TENDER FORM

This tender will be deemed non-compliant and will not be accepted unless a completed Tender Form, Tender Breakdown, Schedule of Details and Unit Rate Schedule is submitted as a part of the tender submitted.

For **Mechanical Services**

At **Skygate Play, Brisbane Airport**

I/We, the undersigned, hereby tender the following sum to complete the works in accordance with the Specification and Drawings

Service	Total Price \$ (Excl. GST)
Mechanical Services	\$
	\$
<u>TOTAL LUMP SUM FIXED PRICE (excl. GST)</u>	<u>\$</u>

19. TENDER FORM B - ITEMISED TENDER FORM

Provision of the following:

Service	Total Price \$ (Excl. GST)
Packaged Air Conditioning Systems	\$
Packaged Outside Air Unit	\$
Drains	\$
Ventilation Fans	\$
Air Distribution System and Ancillaries	\$
Kitchen Exhaust System	\$
Air Filters	\$
Insulation	\$
Electrical & Controls	\$
Painting and Labelling	\$
Testing and Commissioning	\$
Project Management and Engineering	\$
Drafting, O&M Manuals, and associated items	\$
Warranty, Service and Maintenance	\$
Miscellaneous	\$
TOTAL LUMP SUM FIXED PRICE (excl. GST)	\$

20. TENDER FORM C - SCHEDULE OF TECHNICAL DETAILS

The schedule of technical details forms part of the specification and shall be completed in full and returned with the tender.

Provide manufacturer's selection details for tender assessment, including:

- Manufacturer
- As installed capacity.

20.1 Air Conditioning Units

Provide manufacturer's selection data for evaluation.

20.2 Heat Recovery Ventilator

Provide manufacturer's selection data for evaluation.

20.3 Grilles and Diffusers

Grille Type	Manufacturer	Type
Exhaust Grilles		
Air Transfer Grilles		
Return Air Grilles		
Supply Air Registers		

20.4 Electrical Equipment

Description	Details
Starters and Contactors	
Isolating Switches	
Relays	
Fuses	
Circuit Breakers	

20.5 Controls – Air Conditioning

Description	Details
Manufacturer	
Thermostats / Sensors	
A/C Controllers	

20.6 Mechanical Services Control Panel

Description	Details
Manufacturer	

20.7 Insulation

Provide manufacturer's technical details with tender. The following information shall be provided as a minimum: thermal resistivity, thickness, and density.

20.7.1 Ductwork & Fittings

Details	Type 1	Type 2
Manufacturer		
Type		
R-Value		
Thickness		
Density		

20.8 Air Filters

Filter Type	Manufacturer	Type	Performance Rating
Deep Bed			